



```

name: DistressAndMarriage
log: C:\Users\hansg\Documents\working\SHARE data 4\output.smcl
log type: smcl
opened on: 30 Jun 2026, 08:41:51

```

```

1 .
2 . use "C:\Users\hansg\Documents\working\SHARE data 4\dataset.dta"

3 .
4 . keep if country==11 | country==12 | country==13 | country==15 | country==16 | countr
> y==17 | country==18 | country==20 | country==23 | country==28 | country==29 | coun
> try==31 | country==33 | country==34 | country==35 | country==47
(0 observations deleted)

5 .
6 . *panel level id
7 . egen id= group(mergeid)

8 . drop *_f

9 . drop if fdistress==99
(1,718 observations deleted)

10.
11. asdoc, text(\par \qc The Impact of Marriage or Cohabitation on Financial Distress: E
> vidence from SHARE 2015 to 2022 for 15 European Union Countries plus Switzerland) fs
> (12) save(report.doc) replace
(file report.doc not found)
Click to Open File: report.doc

12. asdoc, text(\par \qc Hans Gevers - Junior Research Fellow at the Estonian Business S
> chool https://orcid.org/0009-0001-0249-4142 hans.gevers@ebs.ee) fs(10) save(report.
> doc) append
Click to Open File: report.doc

13.
14. ***variables preparation
15.
16. asdoc, text(\page) save(report.doc) append
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17.
18. generate mst=0

19. replace mst=1 if mstat==3 | mstat==5
(16,540 real changes made)

20. replace mst=2 if mstat==4
(9,068 real changes made)

21. replace mst=3 if mstat==6
(26,111 real changes made)

22. label define msL 0 "Officially with partner" 1 "Away from partner" 2 "Never married"
> 3 "Lost partner"

23. label values mst msL

24.

```

25. spearman mst single

Number of observations = **164,335**
Spearman's rho = **0.8544**

Test of H0: **mst** and **single** are independent
Prob = **0.0000**

26.

27. generate change=0

28. replace change=1 if mst==0
(112,616 real changes made)

29. bysort id: generate changed = change - change[_n-1]
(58,733 missing values generated)

30. replace changed=changed+1
(105,602 real changes made)

31. replace changed=1 if changed==.
(58,733 real changes made)

32. label define chL 0 "No longer officially together" 1 "All other situations" 2 "Newly
> officially together"

33. label values changed chL

34. asdoc tabulate changed, save(report.doc) append

changed	Freq.	Percent	Cum.
No longer officially together	2,269	1.38	1.38
All other situations	159,799	97.24	98.62
Newly officially together	2,267	1.38	100.00
Total	164,335	100.00	

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35.

36. bysort id: generate schanged=single - single[_n-1]
(58,733 missing values generated)

37. replace schanged=0 if schange==.
(58,733 real changes made)

38. replace schanged=schanged+1
(164,335 real changes made)

39. label define schL 0 "Hooked up" 1 "Unchanged" 2 "Back single"

40. label values schanged schL

41. asdoc tabulate schanged, save(report.doc) append

schanged	Freq.	Percent	Cum.
Hooked up	3,215	1.96	1.96
Unchanged	157,896	96.08	98.04
Back single	3,224	1.96	100.00
Total	164,335	100.00	

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42.
43. generate mortg=0

44. replace mortg=1 if mort>0
    (18,830 real changes made)

45. label define mlab 0 "No mortgage" 1 "Mortgage"

46. label values mortg mlab

47.
48. codebook fdistress sphus liab mortg thinc mst changed schanged cjs age yedu gender c
    > ountry, compact

```

Variable	Obs	Unique	Mean	Min	Max	Label
fdistress	164335	4	2.960124	1	4	Household able to make ends meet
sphus	164335	5	3.224219	1	5	Self-perceived health - US scale
liab	164335	3689	2103.205	0	8995232	Financial liabilities
mortg	164335	2	.114583	0	1	
thinc	164335	64291	24731.92	0	7399056	Total household income - Version A
mst	164335	4	.6876746	0	3	
changed	164335	3	.9999878	0	2	
schanged	164335	3	1.000055	0	2	
cjs	164335	7	3.0482	-99	97	Current job situation
age	164335	47	69.58466	50	96	Age
yedu	164335	31	11.11002	0	35	Years of education
gender	164335	2	1.565698	1	2	Gender
country	164335	16	22.78502	11	47	Country identifier

```

49.
50. drop if cjs== -99
    (4 observations deleted)

51. drop if yedu==9997
    (0 observations deleted)

52.
53. winsor2 thinc liab, replace cuts(0 99)

54.
55. codebook fdistress gali sphus liab mortg thinc mst changed schanged cjs age yedu gen
    > der country, compact

```

Variable	Obs	Unique	Mean	Min	Max	Label
fdistress	164331	4	2.960141	1	4	Household able to make ends meet
gali	164331	2	.4949218	0	1	Limitation with activities
sphus	164331	5	3.224212	1	5	Self-perceived health - US scale
liab	164331	3150	1316.461	0	47688.08	Financial liabilities
mortg	164331	2	.1145858	0	1	
thinc	164331	63063	23589.29	0	175014.7	Total household income - Version A
mst	164331	4	.6876183	0	3	
changed	164331	3	.9999878	0	2	
schanged	164331	3	1.000055	0	2	
cjs	164331	6	3.050684	1	97	Current job situation
age	164331	47	69.58427	50	96	Age
yedu	164331	31	11.1101	0	35	Years of education
gender	164331	2	1.565694	1	2	Gender
country	164331	16	22.78489	11	47	Country identifier

```

56.
57. global names liab thinc

58. foreach num of numlist 1(1)2 {
    2.         local name: word `num' of $names
    3.         bysort country: egen mean`name'=mean(`name') if `name'!=0
    4.         generate T=1
    5.         replace T=0 if `name'==0
    6.         replace T=2 if `name'>mean`name'
    7.         drop mean`name'
    8.         drop `name'
    9.         rename T `name'
10. }
(147,992 missing values generated)
(147,992 real changes made)
(5,754 real changes made)
(36,474 missing values generated)
(36,474 real changes made)
(49,908 real changes made)

59. label define liabL 0 "No liabilities" 1 "Below country average liabilities" 2 "Above
> country average liabilities"

60. label values liab liabL

61. label define thL 0 "No income" 1 "Below country average income" 2 "Above country ave
> rage income"

62. label values thinc thL

63.
64. keep fdistress gali sphus liab mortg thinc mst mstat changed schanged cjs age yedu g
> ender country single id Qyear my_wgt

65.
66. asdoc spearman fdistress gali sphus liab mortg thinc single mst changed schanged cjs
> age yedu gender country, save(report.doc) append

```

Number of observations = **164,331**

	fdistr~s	gali	sphus	liab	mortg	thinc	single	mst
> changed schanged	cjs	age	yedu					
fdistress	1.0000							
gali	-0.1639	1.0000						
sphus	-0.2745	0.5450	1.0000					
liab	-0.0548	-0.0142	-0.0400	1.0000				
mortg	0.1458	-0.0774	-0.1657	0.2047	1.0000			
thinc	0.1877	-0.0721	-0.1172	0.1702	0.2194	1.0000		
single	-0.1227	0.0956	0.1066	-0.0489	-0.0842	-0.2044	1.0000	
mst	-0.0997	0.0909	0.1031	-0.0405	-0.0712	-0.1779	0.8544	1.0000
> changed	0.0007	-0.0103	-0.0059	0.0097	0.0054	0.0135	-0.1521	-0.1917
> 1.0000								
> schanged	-0.0078	0.0112	0.0080	-0.0095	-0.0058	-0.0159	0.2171	0.1371
> -0.6970	1.0000							
> cjs	-0.0745	-0.0849	-0.0780	0.1033	0.0834	-0.0259	-0.0685	-0.0621
> 0.0090	-0.0133	1.0000						
> age	0.0270	0.2035	0.2329	-0.1922	-0.1415	-0.0380	0.2133	0.2052
> -0.0278	0.0332	-0.5235	1.0000					
> yedu	0.2472	-0.1161	-0.2180	0.0738	0.1137	0.1397	-0.0696	-0.0772
> 0.0017	-0.0006	0.0285	-0.2095	1.0000				
> gender	-0.0539	0.0415	0.0285	-0.0236	-0.0334	-0.0565	0.2071	0.1989
> 0.0008	-0.0026	0.1268	-0.0131	-0.0594				
> country	-0.2574	0.0677	0.1450	-0.0074	-0.1208	-0.0468	0.0314	0.0356
> -0.0007	0.0029	-0.0319	-0.0504	0.0263				

	gender	country
gender	1.0000	
country	0.0303	1.0000

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67.
68. generate fd=1 if fdistress>2
   (52,522 missing values generated)

69. replace fd=0 if fdistress<3
   (52,522 real changes made)

70. label define fd1  0 "Rather difficult" 1 "Rather easy"

71. label values fd fd1

72.
73. label copy cjs cjsLab

74. label list cjsLab
   cjsLab:
       -99 Not applicable
         1 Retired
         2 Employed or self-employed
         3 Unemployed
         4 Permanently sick
         5 Homemaker
        97 Other

75. replace cjs=6 if cjs==97
   (2,504 real changes made)

76. label define cjsLab 6 "Other", add

77. label values cjs cjsLab

78. tabulate cjs, nolab

```

Current job situation	Freq.	Percent	Cum.
1	110,213	67.07	67.07
2	33,008	20.09	87.15
3	3,244	1.97	89.13
4	4,337	2.64	91.77
5	11,025	6.71	98.48
6	2,504	1.52	100.00
Total	164,331	100.00	

```

79.
80. bysort id: egen Mundlak_OthC_age=mean(age)

81. bysort id: egen Mundlak_OthC_yedu=mean(yedu)

82.
83. *program for calculating Mundlak_Means
84. program define Mundlak_Means
   1.      *syntax varname

```

```

85.      global variab changed schanged mst single sphus gali cjs liab mortg thinc
      2.      foreach V of numlist 1(1)10 {
      3.          local va: word `V' of $variab
      4.          tab `va', gen(_dum_)
      5.          levelsof `va', _local(levels)
      6.          local i = 1
      7.          foreach l of local levels {
      8.              local lbl : label (`va') `l'
      9.              local clean = ustrword("`lbl'",1)
      10.             rename _dum_`i' `va'`clean'
      11.             local ++i
      12.         }
      13.         foreach v of varlist `va'_* {
      14.             if `V'==1 {
      15.                 bysort id: egen Mundlak_C_`v' = mean(`v')
      >             }
      16.             if `V'==2 {
      17.                 bysort id: egen Mundlak_sC_`v' = mean(`v')
      >             }
      18.             if `V'==3 {
      19.                 bysort id: egen Mundlak_Mar_`v' = mean(`v')
      >             }
      20.             if `V'==4 {
      21.                 bysort id: egen Mundlak_Sin_`v' = mean(`v')
      >             }
      22.             if `V'==5 {
      23.                 bysort id: egen Mundlak_Sph_`v' = mean(`v')
      >             }
      24.             if `V'==6 {
      25.                 bysort id: egen Mundlak_Gal_`v' = mean(`v')
      >             }
      26.             if `V'==7 | `V'==8 | `V'==9 | `V'==10 {
      27.                 bysort id: egen Mundlak_Oth_`v' = mean(`v')
      >             }
      28.             }
      29.         }
      30.     }
      31.
      32.
      33.
      34.
      35.
      36.
      37. end

```

86.

87. Mundlak_Means

changed	Freq.	Percent	Cum.
No longer officially together	2,269	1.38	1.38
All other situations	159,795	97.24	98.62
Newly officially together	2,267	1.38	100.00
Total	164,331	100.00	

0 1 2

schanged	Freq.	Percent	Cum.
Hooked up	3,215	1.96	1.96
Unchanged	157,892	96.08	98.04
Back single	3,224	1.96	100.00
Total	164,331	100.00	

0 1 2

mst	Freq.	Percent	Cum.
Officially with partner	112,616	68.53	68.53
Away from partner	16,540	10.07	78.60
Never married	9,068	5.52	84.11
Lost partner	26,107	15.89	100.00
Total	164,331	100.00	

0 1 2 3

Single	Freq.	Percent	Cum.
No	115,784	70.46	70.46
Yes	48,547	29.54	100.00
Total	164,331	100.00	

0 1

Self-perceived health - US scale	Freq.	Percent	Cum.
Excellent	10,131	6.16	6.16
Very good	26,817	16.32	22.48
Good	61,383	37.35	59.84
Fair	48,076	29.26	89.09
Poor	17,924	10.91	100.00
Total	164,331	100.00	

1 2 3 4 5

Limitation with activities	Freq.	Percent	Cum.
Not limited	83,000	50.51	50.51
Limited	81,331	49.49	100.00
Total	164,331	100.00	

0 1

Current job situation	Freq.	Percent	Cum.
Retired	110,213	67.07	67.07
Employed or self-employed	33,008	20.09	87.15
Unemployed	3,244	1.97	89.13
Permanently sick	4,337	2.64	91.77
Homemaker	11,025	6.71	98.48
Other	2,504	1.52	100.00
Total	164,331	100.00	

1 2 3 4 5 6

liab	Freq.	Percent	Cum.
No liabilities	147,992	90.06	90.06
Below country average liabilities	10,585	6.44	96.50
Above country average liabilities	5,754	3.50	100.00
Total	164,331	100.00	

0 1 2

mortg	Freq.	Percent	Cum.
No mortgage	145,501	88.54	88.54
Mortgage	18,830	11.46	100.00
Total	164,331	100.00	

0 1

	thinc	Freq.	Percent	Cum.
No income		36,474	22.20	22.20
Below country average income		77,949	47.43	69.63
Above country average income		49,908	30.37	100.00
Total		164,331	100.00	

0 1 2

88.

89. ***descriptives

90.

91. asdoc, text(\page Part 2: Descriptives) fs(24) save(report.doc) append

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92.

93. xtset id Qyear, yearly

Panel variable: **id** (unbalanced)Time variable: **Qyear, 2015 to 2022**, but with gapsDelta: **1 year**

94. tabulate mstat

	Marital status	Freq.	Percent	Cum.
Married, living with spouse		110,277	67.11	67.11
Registered partnership		2,339	1.42	68.53
Married, not living with spouse		1,835	1.12	69.65
Never married		9,068	5.52	75.16
Divorced		14,705	8.95	84.11
Widowed		26,107	15.89	100.00
Total		164,331	100.00	

95. asdoc xttrans mstat, save(report.doc) append

Marital status		Marital status						To
		1	2	3	4	5	6	
> tal								
	1	96.28	0.01	0.07	0.02	0.29	3.32	100
> .00	2	3.09	93.36	0.00	0.54	0.87	2.15	100
> .00	3	2.04	0.25	90.06	0.08	3.91	3.65	100
> .00	4	1.42	0.21	0.05	97.78	0.21	0.34	100
> .00	5	1.57	0.33	0.06	0.19	97.34	0.50	100
> .00	6	0.55	0.07	0.01	0.03	0.10	99.24	100
	Total	66.11	1.38	1.06	5.46	9.10	16.88	100
> .00								

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96. tabulate single

Single	Freq.	Percent	Cum.
No	115,784	70.46	70.46
Yes	48,547	29.54	100.00
Total	164,331	100.00	

97. asdoc xttrans single, save(report.doc) append

Single	Single		Total
	0	1	
0	94.82	5.18	100.00
1	2.22	97.78	100.00
Total	68.98	31.02	100.00

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98.

99. bysort sphus: asdoc xttrans mst, freq save(report.doc) append

mst	mst				Total
	0	1	2	3	
0	70,873	278	23	2,424	73,598
	96.30	0.38	0.03	3.29	100.00
1	209	10,424	19	91	10,743
	1.95	97.03	0.18	0.85	100.00
2	95	15	5,720	20	5,850
	1.62	0.26	97.78	0.34	100.00
3	95	17	5	15,292	15,409
	0.62	0.11	0.03	99.24	100.00
Total	71,272	10,734	5,767	17,827	105,600
	67.49	10.16	5.46	16.88	100.00

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100 bysort sphus: asdoc xttrans single, freq save(report.doc) append

Single	Single		Total
	0	1	
0	72,192	3,946	76,138
	94.82	5.18	100.00
1	653	28,809	29,462
	2.22	97.78	100.00
Total	72,845	32,755	105,600
	68.98	31.02	100.00

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101

102 bysort gali: asdoc xttrans mst, freq save(report.doc) append

mst	mst				Total
	0	1	2	3	
0	70,873	278	23	2,424	73,598
	96.30	0.38	0.03	3.29	100.00
1	209	10,424	19	91	10,743
	1.95	97.03	0.18	0.85	100.00
2	95	15	5,720	20	5,850
	1.62	0.26	97.78	0.34	100.00

3	95 0.62	17 0.11	5 0.03	15,292 99.24	15,409 100.00
Total	71,272 67.49	10,734 10.16	5,767 5.46	17,827 16.88	105,600 100.00

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103 bysort gali: asdoc xttrans single, freq save(report.doc) append

Single	Single		Total
	0	1	
0	72,192 94.82	3,946 5.18	76,138 100.00
1	653 2.22	28,809 97.78	29,462 100.00
Total	72,845 68.98	32,755 31.02	105,600 100.00

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104

105 asdoc codebook fd fdistress gali sphus liab mortg thinc mst changed schanged cjs age
> yedu gender country single, compact save(report.doc) append

Variable	Obs	Unique	Mean	Min	Max	Label
fd	164331	2	.680389	0	1	
fdistress	164331	4	2.960141	1	4	Household able to make ends meet
gali	164331	2	.4949218	0	1	Limitation with activities
sphus	164331	5	3.224212	1	5	Self-perceived health - US scale
liab	164331	3	.1344421	0	2	
mortg	164331	2	.1145858	0	1	
thinc	164331	3	1.08175	0	2	
mst	164331	4	.6876183	0	3	
changed	164331	3	.9999878	0	2	
schanged	164331	3	1.000055	0	2	
cjs	164331	6	1.664068	1	6	Current job situation
age	164331	47	69.58427	50	96	Age
yedu	164331	31	11.1101	0	35	Years of education
gender	164331	2	1.565694	1	2	Gender
country	164331	16	22.78489	11	47	Country identifier
single	164331	2	.295422	0	1	Single

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106

107 graph bar (count), over(mst, label(angle(90) labsize(small))) over(Qyear, label(angl
> e(0) labsize(small))) scheme(s2color) ytitle("Number of respondents", margin(small))
> ylabel(0(2500)40000,angle(0) labsize(small) grid)

108 graph export descripl.png, replace height(2400)
file **descripl.png** saved as PNG format

109 graph bar (count), over(single, label(angle(90) labsize(small))) over(Qyear, label(a
> ngle(0) labsize(small))) scheme(s2color) ytitle("Number of respondents", margin(smal
> l)) ylabel(0(2500)42500,angle(0) labsize(small) grid)

110 graph export descrip2.png, replace height(2400)
file **descrip2.png** saved as PNG format

111 graph bar (count), over(fdistress, label(angle(90) labsize(small))) over(Qyear, labe
> l(angle(0) labsize(small))) scheme(s2color) ytitle("Number of respondents", margin(s
> mall)) ylabel(0(1000)21000,angle(0) labsize(small) grid)

112 graph export descrip3.png, replace height(2400)
file **descrip3.png** saved as PNG format

113

114 asdoc tabulate fd, save(report.doc) append

fd	Freq.	Percent	Cum.
Rather difficult	52,522	31.96	31.96
Rather easy	111,809	68.04	100.00
Total	164,331	100.00	

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115 asdoc tabulate fdistress, save(report.doc) append

Household able to make ends meet	Freq.	Percent	Cum.
With great difficulty	13,941	8.48	8.48
With some difficulty	38,581	23.48	31.96
Fairly easily	51,896	31.58	63.54
Easily	59,913	36.46	100.00
Total	164,331	100.00	

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116

117 ***regressions

118

119 xtset id Qyear, yearly

Panel variable: **id** (unbalanced)

Time variable: **Qyear, 2015 to 2022**, but with gaps

Delta: **1 year**

120

121 xtlogit fd i.liab i.mortg i.thinc i.mst ibl.changed i.gali i.cjs age yedu i.gender i
> .country

Fitting comparison model:

Iteration 0: Log likelihood = **-102965.85**
Iteration 1: Log likelihood = **-81607.579**
Iteration 2: Log likelihood = **-80582.651**
Iteration 3: Log likelihood = **-80575.525**
Iteration 4: Log likelihood = **-80575.521**

Fitting full model:

tau = **0.0** Log likelihood = **-80575.521**
tau = **0.1** Log likelihood = **-79628.389**
tau = **0.2** Log likelihood = **-78701.205**
tau = **0.3** Log likelihood = **-77812.096**
tau = **0.4** Log likelihood = **-76987.262**
tau = **0.5** Log likelihood = **-76267.67**
tau = **0.6** Log likelihood = **-75724.589**
tau = **0.7** Log likelihood = **-75500.225**
tau = **0.8** Log likelihood = **-75941.21**

```

Iteration 0: Log likelihood = -75500.21
Iteration 1: Log likelihood = -74198.222
Iteration 2: Log likelihood = -74136.095
Iteration 3: Log likelihood = -74135.788
Iteration 4: Log likelihood = -74135.788
Iteration 5: Log likelihood = -74135.788

```

Random-effects logistic regression
Group variable: **id**

Number of obs = **164,331**
Number of groups = **58,731**

Random effects $u_i \sim \text{Gaussian}$

Obs per group:
min = **1**
avg = **2.8**
max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-74135.788**

Wald chi2(34) = **16013.36**
Prob > chi2 = **0.0000**

	fd	Coefficient	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
Below country average liabilities		-.9619655	.0369082	-26.06	0.000	-1.034
> 304						
> -.8896267						
Above country average liabilities		-.8782232	.0501181	-17.52	0.000	-.9764
> 529						
> -.7799935						
	mortg					
Mortgage		-.153727	.0384903	-3.99	0.000	-.2291
> 666						
> -.0782874						
	thinc					
Below country average income		-.1510496	.0211013	-7.16	0.000	-.1924
> 075						
> -.1096918						
Above country average income		.9511016	.0255496	37.23	0.000	.9010
> 253						
> 1.001178						
	mst					
Away from partner		-1.068939	.0401514	-26.62	0.000	-1.147
> 635						
> -.9902442						
Never married		-.5437692	.0521937	-10.42	0.000	-.6460
> 669						
> -.4414715						
Lost partner		-.4991135	.0354004	-14.10	0.000	-.568
> 497						
> -.42973						
	changed					
No longer officially together		.3508623	.0737972	4.75	0.000	.2062
> 225						
> .4955022						
Newly officially together		-.4217286	.0730269	-5.77	0.000	-.5648
> 588						
> -.2785984						
	gali					
Limited		-.6888506	.0200751	-34.31	0.000	-.7281
> 969						
> -.6495042						

		cjs					
> 884	Employed or self-employed		.4392967	.0331171	13.26	0.000	.3743
>	.5042051						
> 166	Unemployed		-1.254203	.0683495	-18.35	0.000	-1.388
>	-1.120241						
> 677	Permanently sick		-.6886198	.0589031	-11.69	0.000	-.8040
>	-.5731718						
> 095	Homemaker		-.4732715	.0429794	-11.01	0.000	-.5575
>	-.3890334						
> 958	Other		-.3504954	.069625	-5.03	0.000	-.486
>	-.2140329						
> 259	age		.0477211	.0016302	29.27	0.000	.0445
>	.0509162						
> 166	yedu		.1179027	.0032072	36.76	0.000	.1116
>	.1241887						
> 497	gender						
>	Female		.0096332	.025655	0.38	0.707	-.0406
> 497							
>	.0599161						
> 936	country						
>	Germany		-.2804179	.0776421	-3.61	0.000	-.4325
> 193	Sweden		-.1167636	.0817136	-1.43	0.153	-.2769
>	.0433921						
> 971	Spain		-2.294824	.0709948	-32.32	0.000	-2.433
>	-2.155677						
> 642	Italy		-3.369346	.0715808	-47.07	0.000	-3.509
>	-3.229051						
> 926	France		-1.592959	.075495	-21.10	0.000	-1.740
>	-1.444991						
> 956	Denmark		.5166948	.0881134	5.86	0.000	.3439
>	.689394						
> 595	Switzerland		.0769773	.0871122	0.88	0.377	-.0937
>	.2477141						
> 518	Belgium		-1.359797	.0712877	-19.07	0.000	-1.499
>	-1.220076						
> 849	Czechia		-1.642493	.0726579	-22.61	0.000	-1.7
>	-1.500086						
> 149	Poland		-3.551916	.0878754	-40.42	0.000	-3.724
>	-3.379684						
> 624	Luxembourg		.0124146	.1059596	0.12	0.907	-.1952
>	.2200916						
> 869	Portugal		-3.527936	.0923145	-38.22	0.000	-3.708
>	-3.347003						
> 927	Slovenia		-3.416156	.0733538	-46.57	0.000	-3.559
>	-3.272385						
> 499	Estonia		-3.280739	.0707972	-46.34	0.000	-3.419
>	-3.141979						

> 761	Croatia	-4.366705	.0837034	-52.17	0.000	-4.530
>		-4.202649				
> 031	_cons	-.8843225	.1357106	-6.52	0.000	-1.15
>		-.6183347				
> 978	/lnsig2u	1.310453	.0196306			1.271
>		1.348929				
> 889	sigma_u	1.925579	.0189002			1.888
>		1.962981				
> 715	rho	.5298659	.0048901			.5202
>		.5394382				

LR test of rho=0: chibar2(01) = 1.3e+04

Prob >= chibar2 = 0.000

122 estimates store RE

123 xtlogit fd i.liab i.mortg i.thinc i.mst ibl.changed i.gali i.cjs age yedu i.gender i
> .country, fe

note: multiple positive outcomes within groups encountered.

note: 43,159 groups (113,419 obs) omitted because of all positive or
all negative outcomes.note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.note: **29.country** omitted because of no within-group variance.note: **31.country** omitted because of no within-group variance.note: **33.country** omitted because of no within-group variance.note: **34.country** omitted because of no within-group variance.note: **35.country** omitted because of no within-group variance.note: **47.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = -18458.706

Iteration 1: Log likelihood = -18101.867

Iteration 2: Log likelihood = -18101.553

Iteration 3: Log likelihood = -18101.553

Conditional fixed-effects logistic regression

Group variable: **id**

Number of obs = 50,912

Number of groups = 15,572

Obs per group:

min = 2

avg = 3.3

max = 4

Log likelihood = -18101.553

LR chi2(18) = 1578.10

Prob > chi2 = 0.0000

		fd	Coefficient	Std. err.	z	P> z	[95%
> con							
> f. interval]							
		liab					
Below country average liabilities			-.4342775	.0444074	-9.78	0.000	-.5213
> 144							
> -.3472405							
Above country average liabilities			-.4915389	.0611089	-8.04	0.000	-.6113
> 101							
> -.3717676							
		mortg					
Mortgage			-.1664527	.0509486	-3.27	0.001	-.2663
> 101							
> -.0665954							
		thinc					
Below country average income			.0034098	.0230145	0.15	0.882	-.0416
> 978							
> .0485174							
Above country average income			.3188759	.0287905	11.08	0.000	.2624
> 475							
> .3753044							
		mst					
Away from partner			-.5001193	.1591488	-3.14	0.002	-.8120
> 453							
> -.1881934							
		Never married	-.3679093	.2936314	-1.25	0.210	-.9434
> 163							
> .2075976							
		Lost partner	-.2144529	.096942	-2.21	0.027	-.4044
> 558							
> -.0244501							
		changed					
No longer officially together			-.0185447	.0947379	-0.20	0.845	-.2042
> 276							
> .1671383							
Newly officially together			.0047409	.0929656	0.05	0.959	-.1774
> 682							
> .18695							
		gali					
Limited			-.300258	.0257266	-11.67	0.000	-.3506
> 812							
> -.2498348							
		cjs					
Employed or self-employed			.2477788	.0475913	5.21	0.000	.1545
> 016							
> .341056							
		Unemployed	-.5278664	.0865733	-6.10	0.000	-.6975
> 469							
> -.3581859							
		Permanently sick	-.09968	.0778279	-1.28	0.200	-.2522
> 199							
> .0528598							
		Homemaker	-.259767	.0667641	-3.89	0.000	-.3906
> 223							
> -.1289118							
		Other	-.0749907	.0849752	-0.88	0.378	-.241
> 539							
> .0915577							
		age	.1060379	.0037148	28.54	0.000	.0987
> 571							
> .1133187							

```

                                yedu |  -.0098688   .0132069   -0.75   0.455   -.0357
> 538
>      .0160162

                                gender
                                Female |      0 (omitted)

                                country
                                Germany |      0 (omitted)
                                Sweden  |      0 (omitted)
                                Spain   |      0 (omitted)
                                Italy    |      0 (omitted)
                                France  |      0 (omitted)
                                Denmark |      0 (omitted)
                                Switzerland |      0 (omitted)
                                Belgium |      0 (omitted)
                                Czechia |      0 (omitted)
                                Poland  |      0 (omitted)
                                Luxembourg |      0 (omitted)
                                Portugal |      0 (omitted)
                                Slovenia |      0 (omitted)
                                Estonia |      0 (omitted)
                                Croatia |      0 (omitted)

```

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125 asdoc hausman FE RE, allegs save(report.doc) append

	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
	(b) FE	(B) RE		
liab				
1	-.4342775	-.9619655	.527688	.0246941
2	-.4915389	-.8782232	.3866843	.0349639
1.mortg	-.1664527	-.153727	-.0127257	.0333804
thinc				
1	.0034098	-.1510496	.1544594	.0091871
2	.3188759	.9511016	-.6322257	.0132707
mst				
1	-.5001193	-1.068939	.5688201	.1540007
2	-.3679093	-.5437692	.1758598	.2889554
3	-.2144529	-.4991135	.2846606	.0902472
changed				
0	-.0185447	.3508623	-.369407	.0594075
2	.0047409	-.4217286	.4264695	.0575296
1.gali	-.300258	-.6888506	.3885926	.0160888
cjs				
2	.2477788	.4392967	-.1915179	.0341787
3	-.5278664	-1.254203	.7263371	.0531346
4	-.09968	-.6886198	.5889398	.0508685
5	-.259767	-.4732715	.2135044	.0510903
6	-.0749907	-.3504954	.2755048	.0487149
age	.1060379	.0477211	.0583168	.003338
yedu	-.0098688	.1179027	-.1277714	.0128115

b = Consistent under H0 and Ha; obtained from **xtlogit**.
 B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

```

chi2(18) = (b-B)'[(V_b-V_B)^(-1)](b-B)
          = 4406.16
Prob > chi2 = 0.0000
(V_b-V_B is not positive definite)
Click to Open File: report.doc

```


126

```
127 xtlogit fd i.liab i.mortg i.thinc i.mst ibl.changed i.sphus i.cjs age yedu i.gender
> i.country
```

Fitting comparison model:

```
Iteration 0: Log likelihood = -102965.85
Iteration 1: Log likelihood = -81259.568
Iteration 2: Log likelihood = -80202.963
Iteration 3: Log likelihood = -80195.089
Iteration 4: Log likelihood = -80195.085
```

Fitting full model:

```
tau = 0.0 Log likelihood = -80195.085
tau = 0.1 Log likelihood = -79258.994
tau = 0.2 Log likelihood = -78343.6
tau = 0.3 Log likelihood = -77467.302
tau = 0.4 Log likelihood = -76656.765
tau = 0.5 Log likelihood = -75953.703
tau = 0.6 Log likelihood = -75430.555
tau = 0.7 Log likelihood = -75231.413
tau = 0.8 Log likelihood = -75705.65
```

```
Iteration 0: Log likelihood = -75231.448
Iteration 1: Log likelihood = -73950.089
Iteration 2: Log likelihood = -73880.465
Iteration 3: Log likelihood = -73880.037
Iteration 4: Log likelihood = -73880.037
Iteration 5: Log likelihood = -73880.037
```

Random-effects logistic regression
Group variable: **id**

Number of obs = **164,331**
Number of groups = **58,731**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.8**
max = **4**

Integration method: **mvaghermite**Integration pts. = **12**Log likelihood = **-73880.037**

Wald chi2(37) = **16258.52**
Prob > chi2 = **0.0000**

	fd	Coefficient	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
Below country average liabilities		-.9515003	.0369177	-25.77	0.000	-1.023
> 858						
> -.8791429						
Above country average liabilities		-.8883925	.0500977	-17.73	0.000	-.9865
> 822						
> -.7902029						
	mortg					
Mortgage		-.1697432	.0384507	-4.41	0.000	-.2451
> 052						
> -.0943813						
	thinc					
Below country average income		-.1607184	.0210888	-7.62	0.000	-.2020
> 516						
> -.1193851						
Above country average income		.9277789	.0255348	36.33	0.000	.8777
> 317						
> .9778261						

> 301	Away from partner	mst	-1.051994	.0399536	-26.33	0.000	-1.130
>	-.973686						
> 442	Never married		-.5253644	.0519294	-10.12	0.000	-.6271
>	-.4235846						
> 704	Lost partner		-.4900159	.0352325	-13.91	0.000	-.5590
>	-.4209614						
> 795	No longer officially together	changed	.3321363	.0737548	4.50	0.000	.1875
>	.4766932						
> 385	Newly officially together		-.4032546	.0730544	-5.52	0.000	-.5464
>	-.2600707						
> 034	Very good	sphus	-.1444503	.0496199	-2.91	0.004	-.2417
>	-.0471971						
> 291	Good		-.5285643	.0469217	-11.26	0.000	-.6205
>	-.4365995						
> 245	Fair		-.9982799	.0484525	-20.60	0.000	-1.093
>	-.9033148						
> 309	Poor		-1.563576	.053946	-28.98	0.000	-1.669
>	-1.457844						
> 701	Employed or self-employed	cjs	.4162414	.0330472	12.60	0.000	.3514
>	.4810127						
> 618	Unemployed		-1.225871	.0682393	-17.96	0.000	-1.359
>	-1.092125						
> 999	Permanently sick		-.5898394	.0589605	-10.00	0.000	-.7053
>	-.4742789						
> 151	Homemaker		-.4606412	.0428446	-10.75	0.000	-.5446
>	-.3766673						
> 239	Other		-.3247629	.0696752	-4.66	0.000	-.4613
>	-.188202						
> 077		age	.0500025	.00163	30.68	0.000	.0468
>	.0531972						
> 325		yedu	.1097989	.0031972	34.34	0.000	.1035
>	.1160652						
> 925	Female	gender	-.0079075	.0255196	-0.31	0.757	-.057
>	.0421101						
> 723	Germany	country	-.1828713	.0773999	-2.36	0.018	-.3345
>	-.0311702						
> 172	Sweden		-.1292735	.0813503	-1.59	0.112	-.2887
>	.0301701						
> 474	Spain		-2.116629	.070466	-30.04	0.000	-2.25
>	-1.978518						

> 769		Italy		-3.213538	.0710376	-45.24	0.000	-3.352
>	-3.074307							
> 314		France		-1.476113	.0751039	-19.65	0.000	-1.623
>	-1.328913							
> 765		Denmark		.4891781	.0878086	5.57	0.000	.3170
>	.6612797							
> 076		Switzerland		.0566978	.0866881	0.65	0.513	-.1132
>	.2266033							
> 987		Belgium		-1.341985	.0709207	-18.92	0.000	-1.480
>	-1.202983							
> 218		Czechia		-1.560513	.0722999	-21.58	0.000	-1.702
>	-1.418807							
> 723		Poland		-3.337371	.0874258	-38.17	0.000	-3.508
>	-3.16602							
> 549		Luxembourg		.0890582	.1056209	0.84	0.399	-.1179
>	.2960714							
> 769		Portugal		-3.274629	.0919095	-35.63	0.000	-3.454
>	-3.09449							
> 472		Slovenia		-3.306543	.0729243	-45.34	0.000	-3.449
>	-3.163614							
> 612		Estonia		-2.98409	.0706756	-42.22	0.000	-3.122
>	-2.845568							
> 347		Croatia		-4.24622	.0832298	-51.02	0.000	-4.409
>	-4.083092							
> 247		_cons		-.6917794	.140893	-4.91	0.000	-.9679
>	-.4156341							
> 745		/lnsig2u		1.290305	.0196741			1.251
>	1.328866							
> 876		sigma_u		1.906278	.0187522			1.869
>	1.943388							
> 196		rho		.524844	.0049064			.5152
>	.5344499							

LR test of rho=0: chibar2(01) = 1.3e+04

Prob >= chibar2 = 0.000

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```

129 xtlogit fd i.liab i.mortg i.thinc i.mst ibl.changed i.sphus i.cjs age yedu i.gender
> i.country, fe
note: multiple positive outcomes within groups encountered.
note: 43,159 groups (113,419 obs) omitted because of all positive or
      all negative outcomes.
note: 2.gender omitted because of no within-group variance.
note: 12.country omitted because of no within-group variance.
note: 13.country omitted because of no within-group variance.
note: 15.country omitted because of no within-group variance.
note: 16.country omitted because of no within-group variance.
note: 17.country omitted because of no within-group variance.
note: 18.country omitted because of no within-group variance.
note: 20.country omitted because of no within-group variance.

```

note: **23.country** omitted because of no within-group variance.
 note: **28.country** omitted because of no within-group variance.
 note: **29.country** omitted because of no within-group variance.
 note: **31.country** omitted because of no within-group variance.
 note: **33.country** omitted because of no within-group variance.
 note: **34.country** omitted because of no within-group variance.
 note: **35.country** omitted because of no within-group variance.
 note: **47.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = **-18452.166**
 Iteration 1: Log likelihood = **-18102.414**
 Iteration 2: Log likelihood = **-18102.103**
 Iteration 3: Log likelihood = **-18102.103**

Conditional fixed-effects logistic regression
 Group variable: **id**

Number of obs = **50,912**
 Number of groups = **15,572**

Obs per group:
 min = **2**
 avg = **3.3**
 max = **4**

Log likelihood = **-18102.103**

LR chi2(21) = **1577.00**
 Prob > chi2 = **0.0000**

	fd	Coefficient	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
Below country average liabilities		-.4296967	.0444233	-9.67	0.000	-.5167
> 647						
> .3426287						
Above country average liabilities		-.4980947	.0610461	-8.16	0.000	-.6177
> 429						
> .3784464						
	mortg					
Mortgage		-.173515	.050924	-3.41	0.001	-.2733
> 243						
> -.0737058						
	thinc					
Below country average income		-.0014042	.0230202	-0.06	0.951	-.046
> 523						
> .0437145						
Above country average income		.3167794	.0287916	11.00	0.000	.2603
> 489						
> .3732098						
	mst					
Away from partner		-.4776128	.1591754	-3.00	0.003	-.7895
> 909						
> -.1656347						
Never married		-.3351511	.294478	-1.14	0.255	-.9123
> 174						
> .2420153						
Lost partner		-.2132177	.096959	-2.20	0.028	-.4032
> 538						
> -.0231815						
	changed					
No longer officially together		-.0214432	.0947375	-0.23	0.821	-.2071
> 254						
> .164239						
Newly officially together		.014622	.0930549	0.16	0.875	-.1677
> 623						
> .1970064						

> 536		sphus Very good	-.0892671	.0615759	-1.45	0.147	-.2099
>	.0314193						
> 526		Good	-.2452619	.0603535	-4.06	0.000	-.3635
>	-.1269713						
> 668		Fair	-.395391	.0634078	-6.24	0.000	-.519
>	-.2711139						
> 978		Poor	-.6511249	.0715181	-9.10	0.000	-.7912
>	-.510952						
> 423		cjs Employed or self-employed	.247222	.0475926	5.19	0.000	.1539
>	.3405018						
> 569		Unemployed	-.5112577	.0866338	-5.90	0.000	-.6810
>	-.3414586						
> 559		Permanently sick	-.0855635	.0780603	-1.10	0.273	-.238
>	.0674319						
> 383		Homemaker	-.2594817	.0667876	-3.89	0.000	-.390
>	-.1285805						
> 634		Other	-.0656828	.0850427	-0.77	0.440	-.2323
>	.1009977						
> 858		age	.1061457	.0037183	28.55	0.000	.098
>	.1134334						
> 976		yedu	-.0101244	.0132518	-0.76	0.445	-.0360
>	.0158487						
		gender					
		Female	0	(omitted)			
		country					
		Germany	0	(omitted)			
		Sweden	0	(omitted)			
		Spain	0	(omitted)			
		Italy	0	(omitted)			
		France	0	(omitted)			
		Denmark	0	(omitted)			
		Switzerland	0	(omitted)			
		Belgium	0	(omitted)			
		Czechia	0	(omitted)			
		Poland	0	(omitted)			
		Luxembourg	0	(omitted)			
		Portugal	0	(omitted)			
		Slovenia	0	(omitted)			
		Estonia	0	(omitted)			
		Croatia	0	(omitted)			

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131 asdoc hausman FE RE, alleqs save(report.doc) append

	Coefficients		(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
	(b) FE	(B) RE		
liab				
1	-.4296967	-.9515003	.5218036	.0247085
2	-.4980947	-.8883925	.3902979	.0348834
1.mortg	-.173515	-.1697432	-.0037718	.0333886
thinc				
1	-.0014042	-.1607184	.1593142	.00923
2	.3167794	.9277789	-.6109996	.0133016
mst				
1	-.4776128	-1.051994	.5743807	.1540796
2	-.3351511	-.5253644	.1902134	.2898632
3	-.2132177	-.4900159	.2767982	.0903311
changed				
0	-.0214432	.3321363	-.3535795	.0594594
2	.014622	-.4032546	.4178766	.0576392
sphus				
2	-.0892671	-.1444503	.0551831	.0364617
3	-.2452619	-.5285643	.2833024	.0379592
4	-.395391	-.9982799	.602889	.0409012
5	-.6511249	-1.563576	.9124514	.0469539
cjs				
2	.247222	.4162414	-.1690194	.0342482
3	-.5112577	-1.225871	.7146137	.0533743
4	-.0855635	-.5898394	.5042759	.0511574
5	-.2594817	-.4606412	.2011594	.051234
6	-.0656828	-.3247629	.2590801	.0487608
age	.1061457	.0500025	.0561432	.003342
yedu	-.0101244	.1097989	-.1199233	.0128604

b = Consistent under H0 and Ha; obtained from **xtlogit**.
B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

chi2(21) = (b-B)'[(V_b-V_B)^(-1)](b-B)
= 4287.66
Prob > chi2 = 0.0000
(V_b-V_B is not positive definite)
Click to Open File: [report.doc](#)

132

133 xtlogit fd i.liab i.mortg i.thinc i.mst ibl.changed i.gali i.cjs age yedu i.gender i
> .country, fe or baselevels

note: multiple positive outcomes within groups encountered.

note: 43,159 groups (113,419 obs) omitted because of all positive or
all negative outcomes.

note: **2.gender** omitted because of no within-group variance.

note: **12.country** omitted because of no within-group variance.

note: **13.country** omitted because of no within-group variance.

note: **15.country** omitted because of no within-group variance.

note: **16.country** omitted because of no within-group variance.

note: **17.country** omitted because of no within-group variance.

note: **18.country** omitted because of no within-group variance.

note: **20.country** omitted because of no within-group variance.

note: **23.country** omitted because of no within-group variance.

note: **28.country** omitted because of no within-group variance.

note: **29.country** omitted because of no within-group variance.

note: **31.country** omitted because of no within-group variance.

note: **33.country** omitted because of no within-group variance.

note: **34.country** omitted because of no within-group variance.

note: **35.country** omitted because of no within-group variance.

note: **47.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = **-18458.706**
 Iteration 1: Log likelihood = **-18101.867**
 Iteration 2: Log likelihood = **-18101.553**
 Iteration 3: Log likelihood = **-18101.553**

Conditional fixed-effects logistic regression
 Group variable: **id**

Number of obs = **50,912**
 Number of groups = **15,572**

Obs per group:
 min = **2**
 avg = **3.3**
 max = **4**

Log likelihood = **-18101.553**

LR chi2(18) = **1578.10**
 Prob > chi2 = **0.0000**

	fd	Odds ratio	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.6477325	.0287641	-9.78	0.000	.5937
> 396						
> .7066353						
Above country average liabilities		.6116844	.0373794	-8.04	0.000	.5426
> 395						
> .6895144						
	mortg					
	No mortgage	1 (base)				
Mortgage		.8466628	.0431362	-3.27	0.001	.7662
> 015						
> .9355737						
	thinc					
	No income	1 (base)				
Below country average income		1.003416	.0230931	0.15	0.882	.9591
> 596						
> 1.049714						
Above country average income		1.375581	.0396037	11.08	0.000	1.300
> 108						
> 1.455434						
	mst					
	Officially with partner	1 (base)				
Away from partner		.6064583	.0965171	-3.14	0.002	.4439
> 491						
> .8284545						
Never married		.6921799	.2032458	-1.25	0.210	.3892
> 956						
> 1.230718						
Lost partner		.8069828	.0782305	-2.21	0.027	.6673
> 399						
> .9758464						
	changed					
No longer officially together		.9816262	.0929972	-0.20	0.845	.8152
> 768						
> 1.181918						
All other situations		1 (base)				
Newly officially together		1.004752	.0934073	0.05	0.959	.8373
> 876						
> 1.205567						
	gali					
	Not limited	1 (base)				
Limited		.7406271	.0190538	-11.67	0.000	.7042
> 082						

>	.7789295						
		cjs					
		Retired	1	(base)			
> 076		Employed or self-employed	1.281176	.0609728	5.21	0.000	1.167
>	1.406432						
		Unemployed	.5898621	.0510663	-6.10	0.000	.4978
> 049							
>	.6989431	Permanently sick	.905127	.0704441	-1.28	0.200	.7770
> 739							
>	1.054282	Homemaker	.7712312	.0514906	-3.89	0.000	.6766
> 357							
>	.8790515	Other	.9277521	.0788359	-0.88	0.378	.7854
> 181							
>	1.09588						
		age	1.111864	.0041303	28.54	0.000	1.103
> 798							
>	1.119989	yedu	.9901798	.0130772	-0.75	0.455	.9648
> 778							
>	1.016145						
		gender					
		Male	1	(base)			
		Female	1	(omitted)			
		country					
		Austria	1	(base)			
		Germany	1	(omitted)			
		Sweden	1	(omitted)			
		Spain	1	(omitted)			
		Italy	1	(omitted)			
		France	1	(omitted)			
		Denmark	1	(omitted)			
		Switzerland	1	(omitted)			
		Belgium	1	(omitted)			
		Czechia	1	(omitted)			
		Poland	1	(omitted)			
		Luxembourg	1	(omitted)			
		Portugal	1	(omitted)			
		Slovenia	1	(omitted)			
		Estonia	1	(omitted)			
		Croatia	1	(omitted)			

```
134 outreg2 using results.xls, excel replace dec(3) alpha(0.01, 0.05, 0.10) symbol(***,
> **, *) ctitle(xtlogitLimFE) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout
```

```
135
```

```
136 xtlogit fd i.liab i.mortg i.thinc i.mst ib1.changed i.gali i.cjs age yedu i.gender i
> .country, or baselevels vce(robust)
```

Fitting comparison model:

```
Iteration 0: Log pseudolikelihood = -102965.85
Iteration 1: Log pseudolikelihood = -81607.579
Iteration 2: Log pseudolikelihood = -80582.651
Iteration 3: Log pseudolikelihood = -80575.525
Iteration 4: Log pseudolikelihood = -80575.521
```

Fitting full model:


```

tau = 0.0    Log pseudolikelihood = -80575.521
tau = 0.1    Log pseudolikelihood = -79628.389
tau = 0.2    Log pseudolikelihood = -78701.205
tau = 0.3    Log pseudolikelihood = -77812.096
tau = 0.4    Log pseudolikelihood = -76987.262
tau = 0.5    Log pseudolikelihood = -76267.67
tau = 0.6    Log pseudolikelihood = -75724.589
tau = 0.7    Log pseudolikelihood = -75500.225
tau = 0.8    Log pseudolikelihood = -75941.21

```

```

Iteration 0: Log pseudolikelihood = -75500.21
Iteration 1: Log pseudolikelihood = -74198.222
Iteration 2: Log pseudolikelihood = -74136.095
Iteration 3: Log pseudolikelihood = -74135.788
Iteration 4: Log pseudolikelihood = -74135.788
Iteration 5: Log pseudolikelihood = -74135.788

```

Calculating robust standard errors ...

Random-effects logistic regression
Group variable: **id**

Random effects $u_i \sim \text{Gaussian}$

Integration method: **mvaghermite**

Log pseudolikelihood = **-74135.788**

Number of obs = **164,331**
Number of groups = **58,731**

Obs per group:
min = **1**
avg = **2.8**
max = **4**

Integration pts. = **12**

Wald chi2(34) = **15621.60**
Prob > chi2 = **0.0000**

(Std. err. adjusted for **58,731**)

> clusters in **id**)

	fd	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.382141	.0141355	-26.01	0.000	.3554
> 165						
> .410875						
Above country average liabilities		.4155205	.0210083	-17.37	0.000	.3763
> 194						
> .4588053						
	mortg					
	No mortgage	1 (base)				
Mortgage		.8575061	.0333562	-3.95	0.000	.7945
> 592						
> .9254397						
	thinc					
	No income	1 (base)				
Below country average income		.859805	.0179125	-7.25	0.000	.8254
> 043						
> .8956395						
Above country average income		2.58856	.0665801	36.98	0.000	2.4
> 613						
> 2.7224						
	mst					
	Officially with partner	1 (base)				
Away from partner		.3433725	.0142738	-25.71	0.000	.3165
> 057						
> .3725199						
Never married		.5805559	.0313179	-10.08	0.000	.5223

> 075							
>	.6453003						
		Lost partner		.6070686	.0217415	-13.94	0.000 .5659
> 172							
>	.6512123						
		changed					
	No longer officially together			1.420292	.1101803	4.52	0.000 1.219
> 958							
>	1.653523						
		All other situations		1	(base)		
	Newly officially together			.655912	.0490162	-5.64	0.000 .5665
> 463							
>	.7593741						
		gali					
		Not limited		1	(base)		
		Limited		.5021529	.010234	-33.80	0.000 .48
> 249							
>	.5226172						
		cjs					
		Retired		1	(base)		
	Employed or self-employed			1.551616	.0527706	12.92	0.000 1.451
> 559							
>	1.658569						
		Unemployed		.285303	.0202823	-17.64	0.000 .2481
> 956							
>	.3279583						
		Permanently sick		.5022688	.0301752	-11.46	0.000 .4464
> 759							
>	.5650339						
		Homemaker		.6229609	.0269481	-10.94	0.000 .5723
> 208							
>	.6780819						
		Other		.704339	.0498149	-4.96	0.000 .6131
> 686							
>	.8090654						
		age		1.048878	.0017321	28.90	0.000 1.045
> 489							
>	1.052278						
		yedu		1.125135	.003741	35.46	0.000 1.117
> 826							
>	1.132491						
		gender					
		Male		1	(base)		
		Female		1.00968	.0259011	0.38	0.707 .9601
> 695							
>	1.061743						
		country					
		Austria		1	(base)		
		Germany		.755468	.0609895	-3.47	0.001 .6449
> 082							
>	.8849814						
		Sweden		.8897955	.0763966	-1.36	0.174 .7519
> 816							
>	1.052866						
		Spain		.1007791	.0072252	-32.01	0.000 .0875
> 679							
>	.1159836						
		Italy		.0344121	.0025214	-45.99	0.000 .0298
> 088							
>	.0397264						
		France		.2033231	.0159961	-20.25	0.000 .174
> 269							
>	.2372213						
		Denmark		1.676477	.1537967	5.63	0.000 1.400
> 587							
>	2.006713						

> 216		Switzerland		1.080018	.0993	0.84	0.402	.9019
>	1.293281							
> 996		Belgium		.2567129	.0193243	-18.06	0.000	.2214
>	.2975244							
> 578		Czechia		.193497	.0143277	-22.18	0.000	.1673
>	.2237188							
> 189		Poland		.0286696	.0025283	-40.28	0.000	.0241
>	.034079							
> 114		Luxembourg		1.012492	.1111961	0.11	0.910	.8164
>	1.255666							
> 099		Portugal		.0293655	.002708	-38.26	0.000	.0245
>	.0351829							
> 496		Slovenia		.0328384	.0024627	-45.55	0.000	.0283
>	.038038							
> 114		Estonia		.0376005	.002731	-45.17	0.000	.0326
>	.0433528							
> 388		Croatia		.012693	.0010827	-51.19	0.000	.0107
>	.0150028							
> 106		_cons		.4129939	.0565995	-6.45	0.000	.3157
>	.5402541							
> 168		/lnsig2u		1.310453	.019783			1.27
>	1.349227							
> 607		sigma_u		1.925579	.0190468			1.888
>	1.963274							
> 197		rho		.5298659	.0049281			.520
>	.5395124							

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

```

137 * /// Mundlak_Gal_* Mundlak_C_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Mar_* ///
>
138 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
> title(xtlogitLim) stnum(replace coef=exp(coef), replace se=coef*se)
  results.xls
  dir : seeout

139 asdoc, text(xtlogitLim) save(report.doc) append
  Click to Open File: report.doc

140 *asdoc test Mundlak_Gal_gali Not Mundlak_Gal_gali Limited Mundlak_C_changed No Mundl
> ak_C_changed All Mundlak_C_changed Newly_Mundlak_Oth_cjs Retired Mundlak_Oth_cjs Emp
> loyed Mundlak_Oth_cjs Unemployed Mundlak_Oth_cjs Permanently Mundlak_Oth_cjs Homemak
> er Mundlak_Oth_cjs Other Mundlak_Oth_liab No Mundlak_Oth_liab Below Mundlak_Oth_liab
> Above Mundlak_Oth_mortg No Mundlak_Oth_mortg Mortgage Mundlak_Oth_thinc No Mundlak
> Oth_thinc Below Mundlak_Oth_thinc Above_Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_M
> ar_mst Officially Mundlak_Mar_mst_Away Mundlak_Mar_mst_Never Mundlak_Mar_mst_Lost, s
> ave(report.doc) append

```

141

```
142 xtlogit fd i.liab i.mortg i.thinc i.mst ibl.changed i.sphus i.cjs age yedu i.gender
> i.country, or baselevels vce(robust)
```

Fitting comparison model:

```
Iteration 0: Log pseudolikelihood = -102965.85
Iteration 1: Log pseudolikelihood = -81259.568
Iteration 2: Log pseudolikelihood = -80202.963
Iteration 3: Log pseudolikelihood = -80195.089
Iteration 4: Log pseudolikelihood = -80195.085
```

Fitting full model:

```
tau = 0.0 Log pseudolikelihood = -80195.085
tau = 0.1 Log pseudolikelihood = -79258.994
tau = 0.2 Log pseudolikelihood = -78343.6
tau = 0.3 Log pseudolikelihood = -77467.302
tau = 0.4 Log pseudolikelihood = -76656.765
tau = 0.5 Log pseudolikelihood = -75953.703
tau = 0.6 Log pseudolikelihood = -75430.555
tau = 0.7 Log pseudolikelihood = -75231.413
tau = 0.8 Log pseudolikelihood = -75705.65
```

```
Iteration 0: Log pseudolikelihood = -75231.448
Iteration 1: Log pseudolikelihood = -73950.089
Iteration 2: Log pseudolikelihood = -73880.465
Iteration 3: Log pseudolikelihood = -73880.037
Iteration 4: Log pseudolikelihood = -73880.037
Iteration 5: Log pseudolikelihood = -73880.037
```

Calculating robust standard errors ...

```
Random-effects logistic regression
Group variable: id
```

```
Number of obs = 164,331
Number of groups = 58,731
```

Random effects u_i ~ Gaussian

```
Obs per group:
    min = 1
    avg = 2.8
    max = 4
```

Integration method: mvaghermite

Integration pts. = 12

Log pseudolikelihood = -73880.037

```
Wald chi2(37) = 15888.61
Prob > chi2 = 0.0000
```

(Std. err. adjusted for 58,731

> clusters in id)

	fd	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.3861612	.0142473	-25.79	0.000	.3592
> 227						
> .4151199						
Above country average liabilities		.4113164	.0208056	-17.56	0.000	.3724
> 943						
> .4541846						
	mortg					
	No mortgage	1 (base)				
Mortgage		.8438815	.0327627	-4.37	0.000	.7820
> 501						
> .9106014						
	thinc					

	No income	1	(base)			
> 731	Below country average income	.8515318	.0177343	-7.72	0.000	.8174
>	.8870096					
> 612	Above country average income	2.528886	.0650173	36.09	0.000	2.404
>	2.659583					
	mst					
	Officially with partner	1	(base)			
> 102	Away from partner	.3492408	.0144142	-25.49	0.000	.322
>	.3786663					
> 731	Never married	.5913398	.0315802	-9.84	0.000	.5325
>	.6565912					
> 136	Lost partner	.6126167	.0218174	-13.76	0.000	.5713
>	.6569057					
	changed					
> 217	No longer officially together	1.393943	.1082006	4.28	0.000	1.197
>	1.622994					
	All other situations	1	(base)			
> 251	Newly officially together	.668142	.049685	-5.42	0.000	.5775
>	.7729771					
	sphus					
	Excellent	1	(base)			
> 052	Very good	.865498	.0430792	-2.90	0.004	.785
>	.9541874					
> 848	Good	.5894506	.0278119	-11.20	0.000	.5373
>	.646561					
> 887	Fair	.3685128	.0180454	-20.39	0.000	.3347
>	.405634					
> 793	Poor	.2093859	.0114078	-28.70	0.000	.1881
>	.2329823					
	cjs					
	Retired	1	(base)			
> 773	Employed or self-employed	1.516252	.051406	12.28	0.000	1.418
>	1.620429					
> 534	Unemployed	.2935018	.0207444	-17.34	0.000	.255
>	.337111					
> 271	Permanently sick	.5544163	.0330554	-9.89	0.000	.493
>	.6231411					
> 149	Homemaker	.630879	.0271686	-10.70	0.000	.5798
>	.6864403					
> 189	Other	.7226987	.0510154	-4.60	0.000	.6293
>	.8299344					
> 878	age	1.051274	.0017353	30.29	0.000	1.047
>	1.05468					
> 836	yedu	1.116054	.0036944	33.17	0.000	1.108
>	1.123318					
	gender					
	Male	1	(base)			
	Female	.9921237	.0253205	-0.31	0.757	.9437

> 172							
>	1.043013						
		country					
		Austria	1	(base)			
> 549		Germany	.8328754	.0667204	-2.28	0.022	.7118
>	.9744702						
		Sweden	.8787336	.0751392	-1.51	0.131	.7431
> 428							
>	1.039064	Spain	.120437	.0085665	-29.76	0.000	.1047
> 648							
>	.1384536	Italy	.0402141	.0029243	-44.19	0.000	.0348
> 723							
>	.0463741	France	.2285241	.0178771	-18.87	0.000	.1960
> 397							
>	.2663914	Denmark	1.630975	.1492482	5.35	0.000	1.363
> 186							
>	1.95137	Switzerland	1.058336	.0963749	0.62	0.534	.8853
> 417							
>	1.265133	Belgium	.2613265	.0195618	-17.93	0.000	.225
> 666							
>	.3026222	Czechia	.2100284	.0155101	-21.13	0.000	.1817
> 267							
>	.2427377	Poland	.0355302	.0031125	-38.10	0.000	.0299
> 249							
>	.0421856	Luxembourg	1.093144	.1190421	0.82	0.413	.8830
> 447							
>	1.353232	Portugal	.0378309	.0034647	-35.76	0.000	.0316
> 149							
>	.0452691	Slovenia	.0366426	.0027328	-44.34	0.000	.0316
> 596							
>	.04241	Estonia	.0505855	.0036664	-41.17	0.000	.0438
> 866							
>	.058307	Croatia	.0143183	.0012143	-50.07	0.000	.0121
> 255							
>	.0169075						
		_cons	.5006844	.0712019	-4.86	0.000	.3788
> 919							
>	.6616262						
		/lnsig2u	1.290305	.0198281			1.251
> 443							
>	1.329168						
		sigma_u	1.906278	.0188989			1.869
> 594							
>	1.943681						
		rho	.524844	.0049448			.5151
> 442							
>	.534525						

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

```

143 * /// Mundlak_Sph_* Mundlak_C_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Mar_* ///
>
144 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(xtlogitHea) stnum(replace coef=exp(coef), replace se=coef*se)
  results.xls
  dir : seeout

145 asdoc, text(xtologitHea) save(report.doc) append
  Click to Open File: report.doc

146 *asdoc test Mundlak_Sph_sphus_Excellent Mundlak_Sph_sphus_Very Mundlak_Sph_sphus_Goo
> d Mundlak_Sph_sphus_Fair Mundlak_Sph_sphus_Poor Mundlak_C_changed No Mundlak_C_chang
> ed All Mundlak_C_changed Newly Mundlak_Oth_cjs_Retired Mundlak_Oth_cjs_Employed Mund
> lak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_Oth_cjs_Homemaker Mundlak
> _Oth_cjs_Other Mundlak_Oth_liab_No Mundlak_Oth_liab_Below Mundlak_Oth_liab_Above Mun
> dlak_Oth_mortg_No Mundlak_Oth_mortg_Mortgage Mundlak_Oth_thinc_No Mundlak_Oth_thinc_
> Below Mundlak_Oth_thinc_Above Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_Mar_mst_Off
> icially Mundlak_Mar_mst_Away Mundlak_Mar_mst_Never Mundlak_Mar_mst_Lost, save(report
> .doc) append

147
148 xtologit fdistress i.liab i.mortg i.thinc i.mst ib1.changed i.gali i.cjs age yedu i.
> gender i.country [pweight=my_wgt], or baselevels vce(robust)

```

Fitting comparison model:

```

Iteration 0: Log likelihood = -4.654e+08
Iteration 1: Log likelihood = -4.142e+08
Iteration 2: Log likelihood = -3.819e+08
Iteration 3: Log likelihood = -3.750e+08
Iteration 4: Log likelihood = -3.749e+08
Iteration 5: Log likelihood = -3.749e+08
Iteration 6: Log likelihood = -3.749e+08

```

Refining starting values:

```
Grid node 0: Log likelihood = -3.608e+08
```

Fitting full model:

```

Iteration 0: Log pseudolikelihood = -3.608e+08
Iteration 1: Log pseudolikelihood = -3.504e+08
Iteration 2: Log pseudolikelihood = -3.497e+08
Iteration 3: Log pseudolikelihood = -3.497e+08
Iteration 4: Log pseudolikelihood = -3.497e+08
Iteration 5: Log pseudolikelihood = -3.497e+08

```

Random-effects ordered logistic regression
Group variable: **id**

Number of obs = **164,286**
Number of groups = **58,701**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.8**
max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log pseudolikelihood = **-3.497e+08**

Wald chi2(34) = **9866.22**
Prob > chi2 = **0.0000**

(Std. err. adjusted for **58,701**)

```
> clusters in id)
```

	fdistress	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1	(base)			
	Below country average liabilities	.3847007	.0220173	-16.69	0.000	.343

		gender					
		Male	1	(base)			
> 9322		Female	1.030492	.042245	0.73	0.464	.950
>	1.116708						
		country					
		Austria	1	(base)			
> 0803		Germany	.6376594	.0540349	-5.31	0.000	.540
>	.7528686						
		Sweden	1.162765	.1090713	1.61	0.108	.967
> 4897							
>	1.397454						
		Spain	.1077728	.011734	-20.46	0.000	.087
> 0629							
>	.1334091						
		Italy	.0345829	.0029486	-39.46	0.000	.029
> 2607							
>	.040873						
		France	.2169032	.0186726	-17.75	0.000	.183
> 2266							
>	.2567694						
		Denmark	2.683453	.2494915	10.62	0.000	2.23
> 6425							
>	3.219835						
		Switzerland	1.268856	.1201226	2.52	0.012	1.05
> 3972							
>	1.527551						
		Belgium	.4008252	.035967	-10.19	0.000	.336
> 1821							
>	.4778982						
		Czechia	.1545612	.0151954	-18.99	0.000	.127
> 4725							
>	.1874066						
		Poland	.0400303	.0039398	-32.70	0.000	.033
> 0076							
>	.0485471						
		Luxembourg	.6719623	.0788027	-3.39	0.001	.533
> 9768							
>	.8456048						
		Portugal	.034873	.0059021	-19.83	0.000	.025
> 0281							
>	.0485904						
		Slovenia	.0515224	.0044761	-34.14	0.000	.043
> 4557							
>	.0610865						
		Estonia	.0459421	.003879	-36.48	0.000	.038
> 9351							
>	.0542101						
		Croatia	.0214176	.0019508	-42.20	0.000	.01
> 7916							
>	.0256035						
		/cut1	-1.845782	.2147393			-2.26
> 6663							
>	-1.424901						
		/cut2	.953413	.2140419			.533
> 8985							
>	1.372927						
		/cut3	3.609056	.2173959			3.18
> 2968							
>	4.035144						
		/sigma2_u	3.368999	.1075592			3.16
> 4647							
>	3.586546						

Note: Estimates are transformed only in the first equation to odds ratios.

```

149 * /// Mundlak_Gal_* Mundlak_C_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Mar_* ///
>
150 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(xtologitLim) stnum(replace coef=exp(coef), replace se=coef*se)
  results.xls
  dir : seeout

151 asdoc, text(xtologitLim) save(report.doc) append
  Click to Open File: report.doc

152 *asdoc test Mundlak_Gal_gali_Not Mundlak_Gal_gali_Limited Mundlak_C_changed No Mundl
> ak_C_changed All Mundlak_C_changed Newly Mundlak_Oth_cjs_Retired Mundlak_Oth_cjs_Emp
> loyed Mundlak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_Oth_cjs_Homemak
> er Mundlak_Oth_cjs_Other Mundlak_Oth_liab_No Mundlak_Oth_liab_Below Mundlak_Oth_liab
> Above Mundlak_Oth_mortg_No Mundlak_Oth_mortg_Mortgage Mundlak_Oth_thinc_No Mundlak_
> Oth_thinc_Below Mundlak_Oth_thinc_Above Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_M
> ar_mst_Officially Mundlak_Mar_mst_Away Mundlak_Mar_mst_Never Mundlak_Mar_mst_Lost,_s
> ave(report.doc) append

153
154 xtologit fdistress i.liab i.mortg i.thinc i.mst ibl.changed i.sphus i.cjs age yedu i
> .gender i.country [pweight=my_wgt], or baselevels vce(robust)

```

Fitting comparison model:

```

Iteration 0: Log likelihood = -4.654e+08
Iteration 1: Log likelihood = -4.123e+08
Iteration 2: Log likelihood = -3.794e+08
Iteration 3: Log likelihood = -3.726e+08
Iteration 4: Log likelihood = -3.724e+08
Iteration 5: Log likelihood = -3.724e+08
Iteration 6: Log likelihood = -3.724e+08

```

Refining starting values:

```
Grid node 0: Log likelihood = -3.588e+08
```

Fitting full model:

```

Iteration 0: Log pseudolikelihood = -3.588e+08
Iteration 1: Log pseudolikelihood = -3.488e+08
Iteration 2: Log pseudolikelihood = -3.483e+08
Iteration 3: Log pseudolikelihood = -3.483e+08
Iteration 4: Log pseudolikelihood = -3.483e+08
Iteration 5: Log pseudolikelihood = -3.483e+08

```

Random-effects ordered logistic regression
Group variable: **id**

Random effects $u_i \sim \text{Gaussian}$

Number of obs = **164,286**
Number of groups = **58,701**

Obs per group:
min = **1**
avg = **2.8**
max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log pseudolikelihood = **-3.483e+08**

Wald chi2(37) = **10252.54**
Prob > chi2 = **0.0000**

(Std. err. adjusted for **58,701**)

```
> clusters in id)
```

	fdistress	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1	(base)			

Below country average liabilities		.3885448	.0220318	-16.67	0.000	.347
> 6763						
> .4342172						
Above country average liabilities		.3971874	.0294199	-12.47	0.000	.343
> 5157						
> .459245						
	mortg					
	No mortgage	1	(base)			
	Mortgage	.7277692	.0411287	-5.62	0.000	.651
> 4625						
> .8130138						
	thinc					
	No income	1	(base)			
Below country average income		.882085	.0291223	-3.80	0.000	.826
> 8138						
> .9410509						
Above country average income		2.514477	.1013524	22.88	0.000	2.32
> 3474						
> 2.721182						
	mst					
	Officially with partner	1	(base)			
	Away from partner	.4138066	.0290717	-12.56	0.000	.360
> 5761						
> .4748954						
	Never married	.6679589	.0583402	-4.62	0.000	.562
> 8659						
> .7926738						
	Lost partner	.7677678	.0432585	-4.69	0.000	.687
> 4966						
> .8574115						
	changed					
No longer officially together		1.28241	.1378917	2.31	0.021	1.03
> 8726						
> 1.583262						
	All other situations	1	(base)			
Newly officially together		.6266953	.0564065	-5.19	0.000	.525
> 3431						
> .7476011						
	sphus					
	Excellent	1	(base)			
	Very good	.9555647	.0825295	-0.53	0.599	.806
> 7596						
> 1.131816						
	Good	.655352	.0530623	-5.22	0.000	.559
> 1841						
> .7680588						
	Fair	.4460473	.036891	-9.76	0.000	.379
> 2984						
> .5245427						
	Poor	.2528888	.0234533	-14.82	0.000	.210
> 8569						
> .3032993						
	cjs					
	Retired	1	(base)			
Employed or self-employed		1.184922	.0579145	3.47	0.001	1.07
> 6679						
> 1.304047						
	Unemployed	.1905278	.0232669	-13.58	0.000	.149
> 9724						
> .2420502						
	Permanently sick	.4946332	.04684	-7.43	0.000	.410
> 8445						
> .59551						
	Homemaker	.652684	.0437596	-6.36	0.000	.57
> 2313						
> .7443415						

		Other		.5584112	.0692859	-4.70	0.000	.437
> 8645								
>	.7121452							
		age		1.048913	.0027816	18.01	0.000	1.04
> 3475								
>	1.054379							
		yedu		1.121902	.0062823	20.54	0.000	1.10
> 9657								
>	1.134283							
		gender						
		Male		1	(base)			
> 0997		Female		1.021775	.0412183	0.53	0.593	.944
>	1.105841							
		country						
		Austria		1	(base)			
> 3881		Germany		.7039626	.0583602	-4.23	0.000	.598
>	.8281639							
		Sweden		1.136057	.1046882	1.38	0.166	.948
> 3343								
>	1.360939							
		Spain		.1249515	.0133616	-19.45	0.000	.101
> 3256								
>	.1540864							
		Italy		.0393742	.0032884	-38.73	0.000	.033
> 4289								
>	.0463769							
		France		.2421947	.020427	-16.81	0.000	.205
> 2926								
>	.2857301							
		Denmark		2.573993	.2358363	10.32	0.000	2.15
> 0889								
>	3.080327							
		Switzerland		1.228173	.1137515	2.22	0.026	1.02
> 4289								
>	1.47264							
		Belgium		.4069881	.0357601	-10.23	0.000	.342
> 6026								
>	.4834737							
		Czechia		.1675664	.0161381	-18.55	0.000	.138
> 7423								
>	.2023788							
		Poland		.0485808	.0047018	-31.25	0.000	.040
> 1868								
>	.0587281							
		Luxembourg		.7213426	.0832059	-2.83	0.005	.575
> 3825								
>	.9043291							
		Portugal		.0442297	.0072377	-19.06	0.000	.032
> 0941								
>	.0609541							
		Slovenia		.0568962	.0048444	-33.67	0.000	.048
> 1514								
>	.0672292							
		Estonia		.0605797	.0050579	-33.58	0.000	.051
> 4351								
>	.0713502							
		Croatia		.0241457	.0021599	-41.63	0.000	.020
> 2626								
>	.0287728							
		/cut1		-1.917455	.226592			-2.36
> 1567								
>	-1.473343							
		/cut2		.8826504	.2253845			.44
> 0905								
>	1.324396							

	/cut3			
> 8823	3.535284	.228093		3.0
> 3.982338				
	/sigma2_u			
> 0765	3.227475	.1035529		3.03
> 3.436953				

Note: Estimates are transformed only in the first equation to odds ratios.

```

155 * /// Mundlak_Sph_* Mundlak_C_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Mar_* ///
>
156 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(xtologitHea) stnum(replace coef=exp(coef), replace se=coef*se)
  results.xls
  dir : seeout

157 asdoc, text(xtologitHea) save(report.doc) append
  Click to Open File: report.doc

158 *asdoc test Mundlak_Sph_sphus_Excellent Mundlak_Sph_sphus_Very Mundlak_Sph_sphus_Goo
> d Mundlak_Sph_sphus_Fair Mundlak_Sph_sphus_Poor Mundlak_C_changed No Mundlak_C_chang
> ed All Mundlak_C_changed Newly Mundlak_Oth_cjs_Retired Mundlak_Oth_cjs_Employed Mund
> lak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_Oth_cjs_Homemaker Mundlak
> _Oth_cjs_Other Mundlak_Oth_liab_No Mundlak_Oth_liab_Below Mundlak_Oth_liab_Above Mun
> dlak_Oth_mortg_No Mundlak_Oth_mortg_Mortgage Mundlak_Oth_thinc_No Mundlak_Oth_thinc
> Below Mundlak_Oth_thinc_Above Mundlak_OthC_age Mundlak_OthC_yedu Mundlak_Mar_mst_Off
> icially Mundlak_Mar_mst_Away Mundlak_Mar_mst_Never Mundlak_Mar_mst_Lost, save(report
> .doc) append

159
160 logit fd i.liab i.mortg i.thinc i.mst ib1.changed i.gali i.cjs age yedu i.gender i.c
> ountry i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)

```

```

Iteration 0: Log pseudolikelihood = -2.184e+08
Iteration 1: Log pseudolikelihood = -1.695e+08
Iteration 2: Log pseudolikelihood = -1.676e+08
Iteration 3: Log pseudolikelihood = -1.676e+08
Iteration 4: Log pseudolikelihood = -1.676e+08

```

Logistic regression

Number of obs = **164,286**
Wald chi2(37) = **8555.74**
Prob > chi2 = **0.0000**
Pseudo R2 = **0.2329**

Log pseudolikelihood = **-1.676e+08**

(Std. err. adjusted for **58,701**)

> clusters in **id**)

	fd	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1	(base)			
Below country average liabilities		.3691311	.0188252	-19.54	0.000	.3340
> 185						
> .4079347						
Above country average liabilities		.3399327	.0265334	-13.82	0.000	.2917
> 108						
> .396126						
	mortg					
	No mortgage	1	(base)			
Mortgage		.7644042	.0473945	-4.33	0.000	.676
> 935						
> .8631755						
	thinc					

		No income	1	(base)			
> 915	Below country average income		1.021535	.0453593	0.48	0.631	.9363
>	1.114421						
> 814	Above country average income		3.430046	.1732135	24.41	0.000	3.106
>	3.786908						
		mst					
	Officially with partner	1	(base)				
> 768	Away from partner	.5545534	.0316429	-10.33	0.000	.4958	
>	.6201733						
> 306	Never married	.7991163	.0563113	-3.18	0.001	.6960	
>	.9174695						
> 366	Lost partner	.9091994	.042593	-2.03	0.042	.8294	
>	.9966327						
		changed					
> 542	No longer officially together	1.235289	.1154404	2.26	0.024	1.028	
>	1.483594						
	All other situations	1	(base)				
> 091	Newly officially together	.7252516	.0720033	-3.24	0.001	.5970	
>	.8810416						
		gali					
	Not limited	1	(base)				
> 902	Limited	.5625966	.0161042	-20.09	0.000	.531	
>	.5950625						
		cjs					
	Retired	1	(base)				
> 691	Employed or self-employed	1.128162	.0553724	2.46	0.014	1.024	
>	1.242082						
> 592	Unemployed	.2112648	.0222518	-14.76	0.000	.1718	
>	.2597056						
> 189	Permanently sick	.4786861	.0400313	-8.81	0.000	.4063	
>	.5639422						
> 925	Homemaker	.7177514	.0407921	-5.84	0.000	.6420	
>	.8023254						
> 067	Other	.6310775	.0639459	-4.54	0.000	.5174	
>	.7697211						
> 377	age	1.028997	.0023625	12.45	0.000	1.024	
>	1.033638						
> 855	yedu	1.075532	.0049599	15.79	0.000	1.065	
>	1.085297						
		gender					
	Male	1	(base)				
> 694	Female	1.019834	.0363781	0.55	0.582	.9509	
>	1.093685						
		country					
	Austria	1	(base)				
> 138	Germany	.7887878	.0629855	-2.97	0.003	.6745	
>	.9224216						
	Sweden	1.137284	.1066274	1.37	0.170	.946	

> 376	1.366704						
>		Spain		.1865454	.0165077	-18.97	0.000 .1568
> 413	.2218751						
>		Italy		.094789	.0071631	-31.18	0.000 .0817
> 397	.1099215						
>		France		.3417964	.0267313	-13.73	0.000 .2932
> 219	.3984176						
>		Denmark		1.601801	.1462615	5.16	0.000 1.339
> 321	1.915721						
>		Switzerland		1.048793	.0976099	0.51	0.609 .8739
> 153	1.258664						
>		Belgium		.4693544	.0372544	-9.53	0.000 .4017
> 333	.5483577						
>		Czechia		.3013519	.0291516	-12.40	0.000 .2493
> 055	.3642637						
>		Poland		.0951912	.008193	-27.33	0.000 .0804
> 146	.1126832						
>		Luxembourg		.9018221	.1025698	-0.91	0.364 .7216
> 199	1.127024						
>		Portugal		.094947	.0121756	-18.36	0.000 .073
> 846	.1220775						
>		Slovenia		.1036519	.0079123	-29.69	0.000 .0892
> 485	.1203799						
>		Estonia		.1230163	.0091782	-28.09	0.000 .1062
> 809	.1423869						
>		Croatia		.0585812	.0047133	-35.27	0.000 .0500
> 349	.0685873						
>							
		Qyear					
		2015					
		2017		1.285282	1 (base) .046245	6.98	0.000 1.197
> 766	1.379193						
>		2020		1.164854	.040502	4.39	0.000 1.088
> 116	1.247004						
>		2022		1.294146	.0480815	6.94	0.000 1.203
> 257	1.3919						
>							
		_cons		.5719827	.1055356	-3.03	0.002 .398
> 409	.8211767						
>							

Note: **_cons** estimates baseline odds.

```
161 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(logitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout
```

```
162
```

```
163 logit fd i.liab i.mortg i.thinc i.mst ib1.changed i.sphus i.cjs age yedu i.gender i.
> country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)
```

```
Iteration 0: Log pseudolikelihood = -2.184e+08
Iteration 1: Log pseudolikelihood = -1.681e+08
Iteration 2: Log pseudolikelihood = -1.661e+08
Iteration 3: Log pseudolikelihood = -1.660e+08
Iteration 4: Log pseudolikelihood = -1.660e+08
```

```
Logistic regression
```

```
Number of obs = 164,286
Wald chi2(40) = 8677.71
Prob > chi2 = 0.0000
Pseudo R2 = 0.2398
```

```
Log pseudolikelihood = -1.660e+08
```

```
(Std. err. adjusted for 58,701
```

```
> clusters in id)
```

	fd	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.3759829	.019018	-19.34	0.000	.3404
> 964						
> .4151679						
Above country average liabilities		.3446411	.0273931	-13.40	0.000	.2949
> 246						
> .4027383						
	mortg					
	No mortgage	1 (base)				
Mortgage		.757214	.047872	-4.40	0.000	.6689
> 669						
> .8571022						
	thinc					
	No income	1 (base)				
Below country average income		1.023781	.0454851	0.53	0.597	.9384
> 025						
> 1.116926						
Above country average income		3.398687	.1729194	24.05	0.000	3.076
> 122						
> 3.755077						
	mst					
	Officially with partner	1 (base)				
Away from partner		.5650847	.0323381	-9.97	0.000	.5051
> 285						
> .6321574						
Never married		.8061098	.0558706	-3.11	0.002	.7037
> 175						
> .9234003						
Lost partner		.9197114	.0429865	-1.79	0.073	.8392
> 033						
> 1.007943						
	changed					
No longer officially together		1.225255	.1161036	2.14	0.032	1.017
> 578						
> 1.475317						
All other situations		1 (base)				
Newly officially together		.7314171	.071935	-3.18	0.001	.6031

> 833							
>	.8869127						
		sphus					
		Excellent	1	(base)			
		Very good	1.054883	.0989094	0.57	0.569	.8777
> 946							
>	1.267698						
		Good	.7142555	.0616185	-3.90	0.000	.6031
> 438							
>	.8458363	Fair	.4678792	.0413313	-8.60	0.000	.3934
> 963							
>	.5563226	Poor	.2871301	.0268782	-13.33	0.000	.2390
> 001							
>	.3449527						
		cjs					
		Retired	1	(base)			
	Employed or self-employed		1.093907	.0538966	1.82	0.068	.9932
> 121							
>	1.204811	Unemployed	.2174028	.0227874	-14.56	0.000	.1770
> 293							
>	.2669839	Permanently sick	.5487047	.0454131	-7.25	0.000	.4665
> 408							
>	.6453387	Homemaker	.7194376	.0406255	-5.83	0.000	.6440
> 612							
>	.8036355	Other	.6381531	.0649422	-4.41	0.000	.5227
> 592							
>	.7790191						
		age	1.032074	.0023783	13.70	0.000	1.027
> 423							
>	1.036746	yedu	1.070707	.0049096	14.90	0.000	1.061
> 128							
>	1.080374						
		gender					
		Male	1	(base)			
		Female	1.012335	.0361259	0.34	0.731	.9439
> 485							
>	1.085675						
		country					
		Austria	1	(base)			
		Germany	.8594042	.0677354	-1.92	0.055	.7363
> 911							
>	1.002966	Sweden	1.129103	.1050341	1.31	0.192	.9409
> 165							
>	1.354927	Spain	.2133186	.0188088	-17.52	0.000	.1794
> 636							
>	.2535601	Italy	.1062843	.0079301	-30.04	0.000	.0918
> 247							
>	.1230208	France	.3756954	.0291304	-12.63	0.000	.3227
> 276							
>	.4373566	Denmark	1.570234	.1427051	4.96	0.000	1.314
> 032							
>	1.876388	Switzerland	1.026885	.0947473	0.29	0.774	.857
> 007							
>	1.230437						

> 048		Belgium		.4675748	.0366399	-9.70	0.000	.4010
>	.5451961							
> 669		Czechia		.314663	.030409	-11.96	0.000	.2603
>	.3802819							
> 434		Poland		.1104135	.0094612	-25.72	0.000	.0933
>	.1306053							
> 675		Luxembourg		.9634147	.1091529	-0.33	0.742	.7715
>	1.202964							
> 944		Portugal		.1172816	.0149889	-16.77	0.000	.0912
>	.1506662							
> 512		Slovenia		.1106962	.0083687	-29.11	0.000	.0954
>	.1283761							
> 238		Estonia		.1565635	.0116459	-24.93	0.000	.1353
>	.1811368							
> 826		Croatia		.0632713	.0050657	-34.48	0.000	.0540
>	.0740213							
		Qyear						
		2015		1	(base)			
> 384		2017		1.285404	.0465204	6.94	0.000	1.197
>	1.379893							
> 937		2020		1.131648	.0399814	3.50	0.000	1.055
>	1.212787							
> 784		2022		1.263806	.047099	6.28	0.000	1.174
>	1.359573							
		_cons		.5593364	.1124444	-2.89	0.004	.3771
> 851								
>	.8294528							

Note: **_cons** estimates baseline odds.

```
164 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(logitHea) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout
```

165

```
166 ologit fdistress i.liab i.mortg i.thinc i.mst ib1.changed i.gali i.cjs age yedu i.ge
> nder i.country i.Qyear
```

```
Iteration 0: Log likelihood = -210570.75
Iteration 1: Log likelihood = -180190.74
Iteration 2: Log likelihood = -179301.35
Iteration 3: Log likelihood = -179299.98
Iteration 4: Log likelihood = -179299.98
```

Ordered logistic regression

Log likelihood = -179299.98

```
Number of obs = 164,331
LR chi2(37) = 62541.54
Prob > chi2 = 0.0000
Pseudo R2 = 0.1485
```

	fdistress	Coefficient	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
Below country average liabilities		-.8169353	.0201352	-40.57	0.000	-.8563
> 995						
> -.7774711						
Above country average liabilities		-.6609427	.027242	-24.26	0.000	-.714
> 336						
> -.6075495						
	mortg					
Mortgage		-.1111117	.0185144	-6.00	0.000	-.1473
> 993						
> -.0748242						
	thinc					
Below country average income		.0254013	.0208074	1.22	0.222	-.0153
> 804						
> .0661829						
Above country average income		.9966506	.0220421	45.22	0.000	.9534
> 488						
> 1.039852						
	mst					
Away from partner		-.5610949	.0165594	-33.88	0.000	-.5935
> 507						
> -.5286392						
Never married		-.2338382	.0214349	-10.91	0.000	-.2758
> 497						
> -.1918267						
Lost partner		-.1910209	.0150096	-12.73	0.000	-.2204
> 391						
> -.1616027						
	changed					
No longer officially together		.1174818	.041067	2.86	0.004	.0369
> 919						
> .1979716						
Newly officially together		-.2157118	.0407299	-5.30	0.000	-.2955
> 411						
> -.1358826						
	gali					
Limited		-.504922	.0100809	-50.09	0.000	-.5246
> 802						
> -.4851639						
	cjs					
Employed or self-employed		.279527	.0157003	17.80	0.000	.248
> 755						
> .310299						
Unemployed		-1.12847	.0361006	-31.26	0.000	-1.199
> 226						
> -1.057714						
Permanently sick		-.6405814	.0311233	-20.58	0.000	-.7015
> 819						
> -.5795808						
Homemaker		-.2983478	.0205082	-14.55	0.000	-.3385
> 432						
> -.2581525						
Other		-.31842	.0393685	-8.09	0.000	-.3955
> 809						
> -.2412592						
	age					
> 049		.0226343	.0007293	31.04	0.000	.0212
> .0240636						

> 812		yedu	.0675916	.0012808	52.77	0.000	.0650
>	.070102						
> 043		gender					
>		Female	.0123735	.0102439	1.21	0.227	-.0077
>	.0324512						
> 177		country					
>		Germany	-.1263078	.0279647	-4.52	0.000	-.1811
>	-.0714979						
> 417		Sweden	.0599451	.0297982	2.01	0.044	.0015
>	.1183485						
> 708		Spain	-1.317916	.0274456	-48.02	0.000	-1.371
>	-1.264124						
> 736		Italy	-2.165324	.0267412	-80.97	0.000	-2.217
>	-2.112912						
> 597		France	-.9105004	.0283471	-32.12	0.000	-.9660
>	-.8549411						
> 191		Denmark	.7387565	.0327238	22.58	0.000	.6746
>	.8028939						
> 507		Switzerland	.0408602	.0315878	1.29	0.196	-.0210
>	.1027712						
> 221		Belgium	-.5803095	.0270478	-21.45	0.000	-.6333
>	-.5272968						
> 508		Czechia	-.9742648	.0271651	-35.86	0.000	-1.027
>	-.9210221						
> 128		Poland	-2.184613	.0339369	-64.37	0.000	-2.251
>	-2.118098						
> 964		Luxembourg	.0256243	.0383276	0.67	0.504	-.0494
>	.100745						
> 925		Portugal	-2.236745	.0383582	-58.31	0.000	-2.311
>	-2.161564						
> 164		Slovenia	-1.958595	.0273314	-71.66	0.000	-2.012
>	-1.905027						
> 736		Estonia	-1.981111	.0263401	-75.21	0.000	-2.032
>	-1.929485						
> 322		Croatia	-2.559313	.0311274	-82.22	0.000	-2.620
>	-2.498305						
> 522		Qyear					
>		2017	.2141184	.019626	10.91	0.000	.1756
>	.2525846						
> 277		2020	-.0221761	.0143633	-1.54	0.123	-.0503
>	.0059756						
> 337		2022	.0853434	.0139338	6.12	0.000	.0580
>	.1126531						
> 055		/cut1	-1.73304	.061233			-1.853
>	-1.613026						
		/cut2	.2765113	.0609657			.1570

```

> 207
>          .396002
                /cut3 |      2.034813      .06114      1.914
> 981
>          2.154645

```

```
167 asdoc estat parallel, save(report.doc) append
```

Tests of proportional odds assumption

```

Number of obs      = 164,331
Number of predictors =    37
Number of outcome levels =    4
Degrees of freedom  =    74

```

Test	chi2	P>chi2
Brant	3029.75	0.000
Likelihood-ratio	3027.4	0.000
Score	3082.39	0.000
Wald	3127.27	0.000
Wolfe-Gould	3040.49	0.000

Click to Open File: [report.doc](#)

```
168
```

```
169 ologit fdistress i.liab i.mortg i.thinc i.mst ib1.changed i.gali i.cjs age yedu i.ge
> nder i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)
```

```

Iteration 0: Log pseudolikelihood = -4.452e+08
Iteration 1: Log pseudolikelihood = -3.770e+08
Iteration 2: Log pseudolikelihood = -3.747e+08
Iteration 3: Log pseudolikelihood = -3.747e+08
Iteration 4: Log pseudolikelihood = -3.747e+08

```

Ordered logistic regression

```

Number of obs = 164,286
Wald chi2(37) = 13690.36
Prob > chi2   = 0.0000
Pseudo R2    = 0.1583

```

Log pseudolikelihood = -3.747e+08

(Std. err. adjusted for 58,701

```
> clusters in id)
```

	fdistress	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.376749	.0164204	-22.40	0.000	.3459
> 018						
> .4103471						
Above country average liabilities		.399924	.0238267	-15.38	0.000	.3558
> 479						
> .4494594						
	mortg					
	No mortgage	1 (base)				
Mortgage		.7686589	.0341775	-5.92	0.000	.7045
> 081						
> .8386512						
	thinc					
	No income	1 (base)				
Below country average income		1.023408	.0365016	0.65	0.517	.9543
> 099						
> 1.09751						
Above country average income		3.24664	.1266097	30.20	0.000	3.007

> 736							
>	3.504521						
		mst					
	Officially with partner	1	(base)				
	Away from partner	.6007771	.0284246	-10.77	0.000	.5475	
> 711							
>	.6591531						
		Never married					
> 834							
>	.9645587						
		Lost partner					
> 315							
>	1.025005						
		changed					
	No longer officially together	1.212242	.0936549	2.49	0.013	1.041	
> 903							
>	1.410428						
	All other situations	1	(base)				
	Newly officially together	.7311048	.0556587	-4.11	0.000	.6297	
> 643							
>	.8487529						
		gali					
	Not limited	1	(base)				
	Limited	.6077204	.0134532	-22.50	0.000	.5819	
> 165							
>	.6346686						
		cjs					
	Retired	1	(base)				
	Employed or self-employed	1.123315	.0417859	3.13	0.002	1.044	
> 331							
>	1.208273						
		Unemployed					
> 864							
>	.2443035						
		Permanently sick					
> 875							
>	.5308696						
		Homemaker					
> 171							
>	.7971528						
		Other					
> 895							
>	.7219602						
		age					
> 718							
>	1.030274						
		yedu					
> 777							
>	1.085914						
		gender					
		Male	1	(base)			
		Female	1.020658	.0277818	0.75	0.453	.9676
> 337							
>	1.076588						
		country					
		Austria	1	(base)			
		Germany	.7822201	.0441144	-4.35	0.000	.7003
> 127							
>	.8737072						
		Sweden					
> 282							
>	1.29992						
		Spain					
> 826							
>	.2559228						

> 281		Italy		.099181	.0054649	-41.94	0.000	.0890
>	.1104918							
> 071		France		.3721176	.0214256	-17.17	0.000	.3324
>	.4165721							
> 897		Denmark		2.103449	.1349547	11.59	0.000	1.854
>	2.385305							
> 673		Switzerland		1.085355	.0701201	1.27	0.205	.9562
>	1.231868							
> 218		Belgium		.5929699	.035498	-8.73	0.000	.5273
>	.6667909							
> 836		Czechia		.2931531	.0194632	-18.48	0.000	.2573
>	.3338936							
> 627		Poland		.1086311	.0069304	-34.79	0.000	.0958
>	.1231003							
> 196		Luxembourg		.8338211	.0654513	-2.32	0.021	.7149
>	.9724975							
> 103		Portugal		.1066953	.0109152	-21.87	0.000	.0873
>	.1303842							
> 684		Slovenia		.1338883	.0076612	-35.14	0.000	.119
>	.1497784							
> 311		Estonia		.1303523	.0071696	-37.04	0.000	.1170
>	.1451897							
> 662		Croatia		.0762875	.0044955	-43.67	0.000	.0679
>	.0856276							
		Qyear						
		2015		1	(base)			
> 258		2017		1.24838	.0352017	7.87	0.000	1.181
>	1.319316							
> 939		2020		1.006198	.0250957	0.25	0.804	.9581
>	1.056606							
> 128		2022		1.091946	.0292324	3.29	0.001	1.036
>	1.15077							
> 989		/cut1		-1.440785	.1526075			-1.73
>	-1.14168							
> 639		/cut2		.5058514	.151374			.2091
>	.8025389							
> 888		/cut3		2.298761	.1519786			2.000
>	2.596633							

Note: Estimates are transformed only in the first equation to odds ratios.

```

170 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *) c
> title(ologitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout

```

```

171

```

```

172 ologit fdistress i.liab i.mortg i.thinc i.mst ib1.changed i.sphus i.cjs age yedu i.g
> ender i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)

```

```

Iteration 0: Log pseudolikelihood = -4.452e+08
Iteration 1: Log pseudolikelihood = -3.747e+08
Iteration 2: Log pseudolikelihood = -3.722e+08
Iteration 3: Log pseudolikelihood = -3.722e+08
Iteration 4: Log pseudolikelihood = -3.722e+08

```

Ordered logistic regression

```

Number of obs = 164,286
Wald chi2(40) = 14239.40
Prob > chi2 = 0.0000
Pseudo R2 = 0.1639

```

Log pseudolikelihood = -3.722e+08

(Std. err. adjusted for 58,701

```

> clusters in id)

```

	fdistress	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.3841305	.0165236	-22.24	0.000	.3530
> 724						
> .4179206						
Above country average liabilities		.406525	.0245459	-14.91	0.000	.3611
> 535						
> .4575964						
	mortg					
	No mortgage	1 (base)				
Mortgage		.761792	.034055	-6.09	0.000	.6978
> 859						
> .8315499						
	thinc					
	No income	1 (base)				
Below country average income		1.023875	.0367285	0.66	0.511	.9543
> 613						
> 1.098453						
Above country average income		3.204162	.1258288	29.65	0.000	2.966
> 794						
> 3.460521						
	mst					
	Officially with partner	1 (base)				
Away from partner		.6077387	.0284746	-10.63	0.000	.5544
> 154						
> .6661905						
Never married		.8631324	.050677	-2.51	0.012	.7693
> 092						
> .9683981						
Lost partner		.9563593	.0385708	-1.11	0.269	.8836
> 727						
> 1.035025						
	changed					
No longer officially together		1.196372	.0921513	2.33	0.020	1.028
> 731						
> 1.391331						
All other situations		1 (base)				
Newly officially together		.7295175	.0556497	-4.13	0.000	.6282

> 081							
>	.8471647						
		sphus					
		Excellent	1	(base)			
> 037		Very good	.9423022	.0642048	-0.87	0.383	.8245
>	1.076931						
		Good	.6691937	.0436646	-6.16	0.000	.5888
> 591							
>	.760488						
		Fair	.4600914	.0309173	-11.55	0.000	.4033
> 155							
>	.5248598						
		Poor	.2777745	.0203327	-17.50	0.000	.24
> 065							
>	.3206262						
		cjs					
		Retired	1	(base)			
> 942		Employed or self-employed	1.091644	.0405772	2.36	0.018	1.014
>	1.174143						
		Unemployed	.2095107	.0198293	-16.51	0.000	.1740
> 378							
>	.2522137						
		Permanently sick	.537571	.0372915	-8.95	0.000	.469
> 232							
>	.6158629						
		Homemaker	.7269132	.0353259	-6.56	0.000	.6608
> 709							
>	.7995552						
		Other	.6103062	.0558264	-5.40	0.000	.510
> 136							
>	.7301457						
		age	1.029665	.0019206	15.67	0.000	1.025
> 908							
>	1.033436						
		yedu	1.072949	.0040445	18.68	0.000	1.065
> 051							
>	1.080905						
		gender					
		Male	1	(base)			
> 801		Female	1.012606	.0274659	0.46	0.644	.9601
>	1.067895						
		country					
		Austria	1	(base)			
> 689		Germany	.8493775	.0474038	-2.93	0.003	.7613
>	.9475593						
		Sweden	1.130597	.072434	1.92	0.055	.9971
> 803							
>	1.281863						
		Spain	.2521643	.0182052	-19.08	0.000	.2188
> 923							
>	.2904937						
		Italy	.1102191	.0060132	-40.42	0.000	.0990
> 416							
>	.122658						
		France	.406039	.0231943	-15.78	0.000	.3630
> 314							
>	.4541416						
		Denmark	2.050274	.1314396	11.20	0.000	1.808
> 185							
>	2.324776						
		Switzerland	1.068313	.068311	1.03	0.301	.9424
> 755							
>	1.210951						

> 931		Belgium		.5945285	.035313	-8.75	0.000	.5291
>	.6679303							
> 675		Czechia		.3068502	.0202808	-17.87	0.000	.2695
>	.3492893							
> 298		Poland		.1253461	.0079871	-32.59	0.000	.1106
>	.14202							
> 027		Luxembourg		.8822776	.0691143	-1.60	0.110	.7567
>	1.028692							
> 823		Portugal		.130374	.0131536	-20.19	0.000	.1069
>	.1588803							
> 799		Slovenia		.1431357	.0081164	-34.28	0.000	.1280
>	.1599612							
> 794		Estonia		.1641392	.0090766	-32.68	0.000	.1472
>	.1829289							
> 921		Croatia		.0828858	.0048573	-42.49	0.000	.0738
>	.0929743							
		Qyear						
		2015						
		2017		1 (base)				
> 201				1.249618	.0354624	7.85	0.000	1.18
>	1.321092							
> 041		2020		.9824226	.0246897	-0.71	0.480	.9352
>	1.032025							
> 991		2022		1.068895	.0287578	2.48	0.013	1.013
>	1.126772							
> 816		/cut1		-1.523818	.1647979			-1.846
>	-1.20082							
> 748		/cut2		.4428117	.1632871			.1227
>	.7628486							
> 354		/cut3		2.253717	.1634537			1.933
>	2.574081							

Note: Estimates are transformed only in the first equation to odds ratios.

```
173 outreg2 using results.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *) c
> title(ologitHea) stnum(replace coef=exp(coef), replace se=coef*se)
results.xls
dir : seeout
```

```
174
175
176
177
```

```
178 xtlogit fd i.liab i.mortg i.thinc i.single ib1.schanged i.gali i.cjs age yedu i.gend
> er i.country
```

Fitting comparison model:

```
Iteration 0: Log likelihood = -102965.85
Iteration 1: Log likelihood = -81687.1
Iteration 2: Log likelihood = -80685.035
Iteration 3: Log likelihood = -80678.238
Iteration 4: Log likelihood = -80678.234
```

Fitting full model:

```
tau = 0.0 Log likelihood = -80678.234
tau = 0.1 Log likelihood = -79721.835
tau = 0.2 Log likelihood = -78785.459
tau = 0.3 Log likelihood = -77887.322
tau = 0.4 Log likelihood = -77053.654
tau = 0.5 Log likelihood = -76325.354
tau = 0.6 Log likelihood = -75773.473
tau = 0.7 Log likelihood = -75539.749
tau = 0.8 Log likelihood = -75969.788
```

```
Iteration 0: Log likelihood = -75539.783
Iteration 1: Log likelihood = -74210.432
Iteration 2: Log likelihood = -74149.99
Iteration 3: Log likelihood = -74149.716
Iteration 4: Log likelihood = -74149.715
Iteration 5: Log likelihood = -74149.715
```

Random-effects logistic regression
Group variable: **id**

Number of obs = **164,331**
Number of groups = **58,731**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.8**
max = **4**

Integration method: **mvaghermite**

Integration pts. = **12**

Log likelihood = **-74149.715**

Wald chi2(32) = **15928.39**
Prob > chi2 = **0.0000**

	fd	Coefficient	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
Below country average liabilities		-.9737463	.0369631	-26.34	0.000	-1.046
> 193						
> -.9012999						
Above country average liabilities		-.8881219	.0501661	-17.70	0.000	-.9864
> 457						
> -.7897981						
	mortg					
Mortgage		-.1537391	.0385478	-3.99	0.000	-.2292
> 913						
> -.0781869						
	thinc					
Below country average income		-.1314347	.0211814	-6.21	0.000	-.1729
> 494						
> -.08992						
Above country average income		.9260398	.0256435	36.11	0.000	.8757
> 795						
> .9763						
	single					

> 223		Yes	-.7906846	.0282341	-28.00	0.000	-.8460
>	-.7353469						
> 989		changed Hooked up	-.4229364	.0611554	-6.92	0.000	-.5427
>	-.3030739						
> 098		Back single	.3843639	.0614573	6.25	0.000	.2639
>	.504818						
> 694		gali Limited	-.6883496	.0201125	-34.22	0.000	-.7277
>	-.6489297						
> 353		cjs Employed or self-employed	.4571713	.0331822	13.78	0.000	.3921
>	.5222072						
> 133		Unemployed	-1.245973	.0684501	-18.20	0.000	-1.380
>	-1.111813						
> 306		Permanently sick	-.685993	.0589998	-11.63	0.000	-.8016
>	-.5703555						
> 741		Homemaker	-.4644759	.0430611	-10.79	0.000	-.5488
>	-.3800776						
> 892		Other	-.3150709	.0698065	-4.51	0.000	-.4518
>	-.1782526						
> 253		age	.0542711	.001605	33.81	0.000	.0511
>	.0574169						
> 784		yedu	.1174701	.0032101	36.59	0.000	.1111
>	.1237618						
> 189		gender Female	.0615793	.0256436	2.40	0.016	.0113
>	.1118398						
> 107		country Germany	-.2685691	.0777778	-3.45	0.001	-.4210
>	-.1161275						
> 997		Sweden	-.1127352	.0819221	-1.38	0.169	-.2732
>	.0478292						
> 844		Spain	-2.252703	.0709918	-31.73	0.000	-2.391
>	-2.113562						
> 762		Italy	-3.317442	.0715209	-46.38	0.000	-3.45
>	-3.177264						
> 031		France	-1.567832	.0756135	-20.73	0.000	-1.716
>	-1.419632						
> 855		Denmark	.50966	.0883049	5.77	0.000	.3365
>	.6827345						
> 434		Switzerland	.0518793	.0873091	0.59	0.552	-.1192
>	.2230019						
> 977		Belgium	-1.350038	.0713989	-18.91	0.000	-1.489
>	-1.210098						
		Czechia	-1.64049	.072711	-22.56	0.000	-1.

> 783							
>	-1.497979						
		Poland		-3.477074	.0878388	-39.58	0.000 -3.649
> 235							
>	-3.304914						
		Luxembourg		.03254	.1060511	0.31	0.759 -.1753
> 164							
>	.2403964						
		Portugal		-3.50165	.0924536	-37.87	0.000 -3.682
> 856							
>	-3.320445						
		Slovenia		-3.3503	.0732343	-45.75	0.000 -3.493
> 837							
>	-3.206764						
		Estonia		-3.277853	.07091	-46.23	0.000 -3.416
> 834							
>	-3.138872						
		Croatia		-4.305203	.0836273	-51.48	0.000 -4.469
> 109							
>	-4.141296						
		_cons		-1.373422	.1336063	-10.28	0.000 -1.635
> 285							
>	-1.111558						
		/lnsig2u		1.321209	.0195811		1.282
> 831							
>	1.359587						
		sigma_u		1.935962	.0189542		1.899
> 167							
>	1.97347						
		rho		.5325443	.0048745		.5229
> 796							
>	.5420851						

LR test of rho=0: chibar2(01) = 1.3e+04

Prob >= chibar2 = 0.000

179 estimates store RE

180 xtlogit fd i.liab i.mortg i.thinc i.single ib1.schanged i.gali i.cjs age yedu i.gend

> er i.country, fe

note: multiple positive outcomes within groups encountered.

note: 43,159 groups (113,419 obs) omitted because of all positive or
all negative outcomes.note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.note: **29.country** omitted because of no within-group variance.note: **31.country** omitted because of no within-group variance.note: **33.country** omitted because of no within-group variance.note: **34.country** omitted because of no within-group variance.note: **35.country** omitted because of no within-group variance.note: **47.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = -18457.074

Iteration 1: Log likelihood = -18075.946

Iteration 2: Log likelihood = -18075.562

Iteration 3: Log likelihood = -18075.562

Conditional fixed-effects logistic regression
Group variable: **id**

Number of obs = **50,912**
Number of groups = **15,572**

Obs per group:
min = **2**
avg = **3.3**
max = **4**

Log likelihood = **-18075.562**

LR chi2(16) = **1630.08**
Prob > chi2 = **0.0000**

	fd	Coefficient	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
Below country average liabilities		-.4356656	.0444597	-9.80	0.000	-.5228
> 051						
> -.3485261						
Above country average liabilities		-.4946196	.061154	-8.09	0.000	-.6144
> 792						
> -.3747599						
	mortg					
Mortgage		-.1655159	.0510206	-3.24	0.001	-.2655
> 144						
> -.0655174						
	thinc					
Below country average income		.0130483	.0230702	0.57	0.572	-.0321
> 685						
> .0582651						
Above country average income		.307532	.0288348	10.67	0.000	.2510
> 168						
> .3640472						
	single					
Yes		-.470292	.0792561	-5.93	0.000	-.6256
> 311						
> -.3149529						
	schanged					
Hooked up		-.040878	.0761596	-0.54	0.591	-.190
> 148						
> .1083921						
Back single		.0113853	.0803552	0.14	0.887	-.146
> 108						
> .1688786						
	gali					
Limited		-.2991741	.0257466	-11.62	0.000	-.3496
> 364						
> -.2487117						
	cjs					
Employed or self-employed		.2579407	.0476118	5.42	0.000	.1646
> 232						
> .3512581						
Unemployed		-.5215903	.0866713	-6.02	0.000	-.6914
> 629						
> -.3517177						
Permanently sick		-.0976513	.0778635	-1.25	0.210	-.250
> 261						
> .0549584						
Homemaker		-.2625702	.0668075	-3.93	0.000	-.3935
> 106						
> -.1316299						
Other		-.0721149	.0851042	-0.85	0.397	-.2389
> 162						

```

> .0946863
> 808
> .1183664
> 623
> .0158956

```

	age				
	.1110236	.0037464	29.63	0.000	.1036
	yedu	-.0099834	.0132038	-0.76	0.450
					-.0358
gender					
Female	0	(omitted)			
country					
Germany	0	(omitted)			
Sweden	0	(omitted)			
Spain	0	(omitted)			
Italy	0	(omitted)			
France	0	(omitted)			
Denmark	0	(omitted)			
Switzerland	0	(omitted)			
Belgium	0	(omitted)			
Czechia	0	(omitted)			
Poland	0	(omitted)			
Luxembourg	0	(omitted)			
Portugal	0	(omitted)			
Slovenia	0	(omitted)			
Estonia	0	(omitted)			
Croatia	0	(omitted)			

181 estimates store FE

182 asdoc hausman FE RE, alleqs save(report.doc) append

	Coefficients			
	(b) FE	(B) RE	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
liab				
1	-.4356656	-.9737463	.5380807	.0247061
2	-.4946196	-.8881219	.3935023	.0349739
1.mortg	-.1655159	-.1537391	-.0117768	.0334241
thinc				
1	.0130483	-.1314347	.144483	.0091425
2	.307532	.9260398	-.6185078	.0131856
1.single	-.470292	-.7906846	.3203926	.0740565
schanged				
0	-.040878	-.4229364	.3820584	.0453904
2	.0113853	.3843639	-.3729786	.0517684
1.gali	-.2991741	-.6883496	.3891755	.0160739
cjs				
2	.2579407	.4571713	-.1992306	.0341442
3	-.5215903	-1.245973	.7243826	.0531648
4	-.0976513	-.685993	.5883417	.0508109
5	-.2625702	-.4644759	.2019056	.0510782
6	-.0721149	-.3150709	.242956	.0486804
age	.1110236	.0542711	.0567525	.0033852
yedu	-.0099834	.1174701	-.1274535	.0128076

b = Consistent under H0 and Ha; obtained from **xtlogit**.
B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

chi2(16) = (b-B)'[(V_b-V_B)^(-1)](b-B)
= **4257.64**
Prob > chi2 = **0.0000**
(V_b-V_B is not positive definite)
Click to Open File: [report.doc](#)

183

```
184 xtlogit fd i.liab i.mortg i.thinc i.single ib1.schanged i.sphus i.cjs age yedu i.gen
> der i.country
```

Fitting comparison model:

```
Iteration 0: Log likelihood = -102965.85
Iteration 1: Log likelihood = -81332.109
Iteration 2: Log likelihood = -80297.411
Iteration 3: Log likelihood = -80289.905
Iteration 4: Log likelihood = -80289.901
```

Fitting full model:

```
tau = 0.0 Log likelihood = -80289.901
tau = 0.1 Log likelihood = -79344.767
tau = 0.2 Log likelihood = -78420.412
tau = 0.3 Log likelihood = -77535.33
tau = 0.4 Log likelihood = -76716.228
tau = 0.5 Log likelihood = -76004.762
tau = 0.6 Log likelihood = -75473.172
tau = 0.7 Log likelihood = -75265.093
tau = 0.8 Log likelihood = -75728.933
```

```
Iteration 0: Log likelihood = -75265.206
Iteration 1: Log likelihood = -73957.456
Iteration 2: Log likelihood = -73889.637
Iteration 3: Log likelihood = -73889.249
Iteration 4: Log likelihood = -73889.249
Iteration 5: Log likelihood = -73889.249
```

Random-effects logistic regression
Group variable: **id**

Number of obs = **164,331**
Number of groups = **58,731**

Random effects u_i ~ Gaussian

Obs per group:
min = **1**
avg = **2.8**
max = **4**

Integration method: **mvaghermite**Integration pts. = **12**Log likelihood = **-73889.249**

Wald chi2(35) = **16174.32**
Prob > chi2 = **0.0000**

	fd	Coefficient	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
Below country average liabilities		-.9631406	.0369725	-26.05	0.000	-1.035
> 605						
> -.8906758						
Above country average liabilities		-.8986807	.0501454	-17.92	0.000	-.9969
> 638						
> -.8003975						
	mortg					
Mortgage		-.1699316	.0385078	-4.41	0.000	-.2454
> 055						
> -.0944578						
	thinc					
Below country average income		-.1409291	.0211689	-6.66	0.000	-.1824
> 194						
> -.0994388						
Above country average income		.9025656	.0256282	35.22	0.000	.8523
> 351						
> .952796						

> 931	single	Yes	-.7806256	.0280962	-27.78	0.000	-.8356
>							
>			-.7255581				
> 086	schanged	Hooked up	-.4080741	.0611922	-6.67	0.000	-.5280
>							
>			-.2881396				
> 037	Back	single	.3653399	.0613802	5.95	0.000	.245
>							
>			.4856428				
> 989	sphus	Very good	-.1420047	.0496918	-2.86	0.004	-.2393
>							
>			-.0446105				
> 922	Good		-.529878	.0469979	-11.27	0.000	-.6219
>							
>			-.4377639				
> 387	Fair		-.9972622	.0485342	-20.55	0.000	-1.092
>							
>			-.902137				
> 078	Poor		-1.569143	.0540492	-29.03	0.000	-1.675
>							
>			-1.463209				
> 945	cjs	Employed or self-employed	.4342898	.0331105	13.12	0.000	.3693
>							
>			.4991851				
> 695	Unemployed		-1.216757	.0683373	-17.81	0.000	-1.350
>							
>			-1.082818				
> 299	Permanently sick		-.5850659	.0590644	-9.91	0.000	-.7008
>							
>			-.4693019				
> 999	Homemaker		-.4526684	.0429251	-10.55	0.000	-.5367
>							
>			-.3685368				
> 592	Other		-.2889295	.0698634	-4.14	0.000	-.4258
>							
>			-.1519998				
> 857	age		.0565316	.0016051	35.22	0.000	.0533
>							
>			.0596774				
> 639	yedu		.1093354	.0031998	34.17	0.000	.1030
>							
>			.1156069				
> 496	gender	Female	.0437427	.0255067	1.71	0.086	-.0062
>							
>			.093735				
> 347	country	Germany	-.1712722	.0775333	-2.21	0.027	-.3232
>							
>			-.0193097				
> 999	Sweden		-.1259531	.081556	-1.54	0.122	-.2857
>							
>			.0338936				
> 585	Spain		-2.075479	.0704632	-29.45	0.000	-2.213
>							
>			-1.937374				
> 354	Italy		-3.163236	.07098	-44.57	0.000	-3.302
>							
>			-3.024118				
> 361	France		-1.450933	.0752197	-19.29	0.000	-1.598
>							
>			-1.303505				

> 947		Denmark		.4816666	.0879975	5.47	0.000	.3091
>	.6541385							
> 895		Switzerland		.0313905	.0868819	0.36	0.718	-.138
>	.2016759							
> 719		Belgium		-1.332503	.0710298	-18.76	0.000	-1.471
>	-1.193288							
> 037		Czechia		-1.558568	.072349	-21.54	0.000	-1.70
>	-1.416767							
> 769		Poland		-3.263491	.0873881	-37.34	0.000	-3.434
>	-3.092214							
> 288		Luxembourg		.1082463	.1057035	1.02	0.306	-.0989
>	.3154213							
> 558		Portugal		-3.24915	.0920469	-35.30	0.000	-3.429
>	-3.068741							
> 482		Slovenia		-3.241787	.072805	-44.53	0.000	-3.384
>	-3.099092							
> 367		Estonia		-2.980631	.0707849	-42.11	0.000	-3.119
>	-2.841895							
> 757		Croatia		-4.185781	.0831527	-50.34	0.000	-4.348
>	-4.022804							
> 571		_cons		-1.176491	.1388186	-8.48	0.000	-1.448
>	-.9044118							
> 383		/lnsig2u		1.300849	.0196259			1.262
>	1.339315							
> 849		sigma_u		1.916354	.018805			1.879
>	1.953568							
> 762		rho		.5274726	.0048917			.5178
>	.5370488							

LR test of rho=0: chibar2(01) = 1.3e+04

Prob >= chibar2 = 0.000

185 estimates store RE

186 xtlogit fd i.liab i.mortg i.thinc i.single ib1.schanged i.sphus i.cjs age yedu i.gen

> der i.country, fe

note: multiple positive outcomes within groups encountered.

note: 43,159 groups (113,419 obs) omitted because of all positive or
all negative outcomes.note: **2.gender** omitted because of no within-group variance.note: **12.country** omitted because of no within-group variance.note: **13.country** omitted because of no within-group variance.note: **15.country** omitted because of no within-group variance.note: **16.country** omitted because of no within-group variance.note: **17.country** omitted because of no within-group variance.note: **18.country** omitted because of no within-group variance.note: **20.country** omitted because of no within-group variance.note: **23.country** omitted because of no within-group variance.note: **28.country** omitted because of no within-group variance.note: **29.country** omitted because of no within-group variance.note: **31.country** omitted because of no within-group variance.note: **33.country** omitted because of no within-group variance.note: **34.country** omitted because of no within-group variance.

note: **35.country** omitted because of no within-group variance.
 note: **47.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = **-18449.897**
 Iteration 1: Log likelihood = **-18075.736**
 Iteration 2: Log likelihood = **-18075.358**
 Iteration 3: Log likelihood = **-18075.358**

Conditional fixed-effects logistic regression
 Group variable: **id**

Number of obs = **50,912**
 Number of groups = **15,572**

Obs per group:
 min = **2**
 avg = **3.3**
 max = **4**

Log likelihood = **-18075.358**

LR chi2(19) = **1630.49**
 Prob > chi2 = **0.0000**

	fd	Coefficient	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
Below country average liabilities		-.4313243	.0444781	-9.70	0.000	-.5184
> 999						
> -.3441487						
Above country average liabilities		-.5009697	.0610942	-8.20	0.000	-.6207
> 121						
> -.3812272						
	mortg					
Mortgage		-.1724009	.0509984	-3.38	0.001	-.2723
> 559						
> -.0724458						
	thinc					
Below country average income		.0083484	.0230761	0.36	0.718	-.0368
> 799						
> .0535767						
Above country average income		.3054952	.0288363	10.59	0.000	.2489
> 771						
> .3620134						
	single					
Yes		-.4726811	.0793291	-5.96	0.000	-.6281
> 633						
> -.3171988						
	schanged					
Hooked up		-.0380972	.0761839	-0.50	0.617	-.1874
> 149						
> .1112205						
Back single		.0111656	.0803657	0.14	0.890	-.1463
> 482						
> .1686795						
	sphus					
Very good		-.0853651	.0616133	-1.39	0.166	-.206
> 125						
> .0353947						
Good		-.2441149	.060381	-4.04	0.000	-.3624
> 594						
> -.1257704						
Fair		-.3940791	.0634434	-6.21	0.000	-.5184
> 259						
> -.2697322						
Poor		-.6499668	.0715683	-9.08	0.000	-.7902
> 381						

>	-.5096955					
		cjs				
> 297	Employed or self-employed		.2572508	.0476137	5.40	0.000 .1639
>	.3505719					
> 387	Unemployed		-.5053548	.0867281	-5.83	0.000 -.6753
>	-.3353709					
> 845	Permanently sick		-.0831098	.0781007	-1.06	0.287 -.2361
>	.0699648					
> 784	Homemaker		-.2621007	.0668266	-3.92	0.000 -.3930
>	-.1311229					
> 649	Other		-.0627505	.085162	-0.74	0.461 -.2296
>	.104164					
		age				
> 024			.1111516	.0037497	29.64	0.000 .1038
>	.1185008					
		yedu				
> 153			-.0103525	.0132466	-0.78	0.434 -.0363
>	.0156103					
		gender				
		Female	0	(omitted)		
		country				
		Germany	0	(omitted)		
		Sweden	0	(omitted)		
		Spain	0	(omitted)		
		Italy	0	(omitted)		
		France	0	(omitted)		
		Denmark	0	(omitted)		
		Switzerland	0	(omitted)		
		Belgium	0	(omitted)		
		Czechia	0	(omitted)		
		Poland	0	(omitted)		
		Luxembourg	0	(omitted)		
		Portugal	0	(omitted)		
		Slovenia	0	(omitted)		
		Estonia	0	(omitted)		
		Croatia	0	(omitted)		

187 estimates store FE

188 asdoc hausman FE RE, alleqs save(report.doc) append

	Coefficients			
	(b) FE	(B) RE	(b-B) Difference	sqrt(diag(V_b-V_B)) Std. err.
liab				
1	-.4313243	-.9631406	.5318163	.0247253
2	-.5009697	-.8986807	.397711	.034899
1.mortg	-.1724009	-.1699316	-.0024693	.0334363
thinc				
1	.0083484	-.1409291	.1492775	.009186
2	.3054952	.9025656	-.5970703	.0132184
1.single	-.4726811	-.7806256	.3079445	.074187
schanged				
0	-.0380972	-.4080741	.3699769	.0453817
2	.0111656	.3653399	-.3541743	.051876
sphus				
2	-.0853651	-.1420047	.0566396	.0364269
3	-.2441149	-.529878	.2857631	.0379086
4	-.3940791	-.9972622	.6031831	.0408596
5	-.6499668	-1.569143	.9191764	.0469117

cjs				
2	.2572508	.4342898	-.177039	.0342164
3	-.5053548	-1.216757	.7114018	.053402
4	-.0831098	-.5850659	.5019561	.0510992
5	-.2621007	-.4526684	.1905677	.0512175
6	-.0627505	-.2889295	.226179	.0486998
age	.1111516	.0565316	.05462	.0033888
yedu	-.0103525	.1093354	-.1196879	.0128543

b = Consistent under H0 and Ha; obtained from **xtlogit**.
B = Inconsistent under Ha, efficient under H0; obtained from **xtlogit**.

Test of H0: Difference in coefficients not systematic

chi2(19) = (b-B)'[(V_b-V_B)^(-1)](b-B)
= **4149.54**

Prob > chi2 = **0.0000**

(V b-V B is not positive definite)

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189

190 xtlogit fd i.liab i.mortg i.thinc i.single ib1.schanged i.gali i.cjs age yedu i.gend

> er i.country, fe or baselevels

note: multiple positive outcomes within groups encountered.

note: 43,159 groups (113,419 obs) omitted because of all positive or
all negative outcomes.

note: **2.gender** omitted because of no within-group variance.

note: **12.country** omitted because of no within-group variance.

note: **13.country** omitted because of no within-group variance.

note: **15.country** omitted because of no within-group variance.

note: **16.country** omitted because of no within-group variance.

note: **17.country** omitted because of no within-group variance.

note: **18.country** omitted because of no within-group variance.

note: **20.country** omitted because of no within-group variance.

note: **23.country** omitted because of no within-group variance.

note: **28.country** omitted because of no within-group variance.

note: **29.country** omitted because of no within-group variance.

note: **31.country** omitted because of no within-group variance.

note: **33.country** omitted because of no within-group variance.

note: **34.country** omitted because of no within-group variance.

note: **35.country** omitted because of no within-group variance.

note: **47.country** omitted because of no within-group variance.

Iteration 0: Log likelihood = **-18457.074**

Iteration 1: Log likelihood = **-18075.946**

Iteration 2: Log likelihood = **-18075.562**

Iteration 3: Log likelihood = **-18075.562**

Conditional fixed-effects logistic regression

Group variable: **id**

Number of obs = **50,912**

Number of groups = **15,572**

Obs per group:

min = **2**

avg = **3.3**

max = **4**

LR chi2(16) = **1630.08**

Prob > chi2 = **0.0000**

Log likelihood = **-18075.562**

	fd	Odds ratio	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1	(base)			
Below country average liabilities		.646834	.0287581	-9.80	0.000	.5928
> 552						
> .7057275						
Above country average liabilities		.6098028	.0372919	-8.09	0.000	.5409

> 225							
>	.6874543						
		mortg					
		No mortgage	1	(base)			
> 114		Mortgage	.8474564	.0432377	-3.24	0.001	.7668
>	.9365827						
		thinc					
		No income	1	(base)			
> 434		Below country average income	1.013134	.0233732	0.57	0.572	.9683
>	1.059996						
> 332		Above country average income	1.360064	.0392172	10.67	0.000	1.285
>	1.439142						
		single					
		No	1	(base)			
> 237		Yes	.6248198	.0495208	-5.93	0.000	.5349
>	.7298233						
		schanged					
> 368		Hooked up	.9599463	.0731091	-0.54	0.591	.8268
>	1.114485						
		Unchanged	1	(base)			
> 643		Back single	1.01145	.0812753	0.14	0.887	.8640
>	1.183976						
		gali					
		Not limited	1	(base)			
> 444		Limited	.7414303	.0190893	-11.62	0.000	.7049
>	.7798047						
		cjs					
		Retired	1	(base)			
> 949		Employed or self-employed	1.294262	.0616222	5.42	0.000	1.178
>	1.420854						
		Unemployed	.5935758	.051446	-6.02	0.000	.5008
> 428							
>	.7034787						
> 975		Permanently sick	.9069651	.0706195	-1.25	0.210	.7785
>	1.056497						
		Homemaker	.7690723	.0513798	-3.93	0.000	.6746
> 842							
>	.8766654						
		Other	.930424	.079183	-0.85	0.397	.7874
> 809							
>	1.099314						
		age	1.117421	.0041863	29.63	0.000	1.109
> 246							
>	1.125656						
		yedu	.9900663	.0130726	-0.76	0.450	.9647
> 732							
>	1.016023						
		gender					
		Male	1	(base)			
		Female	1	(omitted)			
		country					
		Austria	1	(base)			
		Germany	1	(omitted)			
		Sweden	1	(omitted)			

Spain	1	(omitted)
Italy	1	(omitted)
France	1	(omitted)
Denmark	1	(omitted)
Switzerland	1	(omitted)
Belgium	1	(omitted)
Czechia	1	(omitted)
Poland	1	(omitted)
Luxembourg	1	(omitted)
Portugal	1	(omitted)
Slovenia	1	(omitted)
Estonia	1	(omitted)
Croatia	1	(omitted)

```
191 outreg2 using results2.xls, excel replace dec(3) alpha(0.01, 0.05, 0.10) symbol(**,
> **, *) ctitle(xtlogitLimFE) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout
```

```
192
```

```
193 xtlogit fd i.liab i.mortg i.thinc i.single ib1.schanged i.gali i.cjs age yedu i.gend
> er i.country, or baselevels vce(robust)
```

Fitting comparison model:

```
Iteration 0: Log pseudolikelihood = -102965.85
Iteration 1: Log pseudolikelihood = -81687.1
Iteration 2: Log pseudolikelihood = -80685.035
Iteration 3: Log pseudolikelihood = -80678.238
Iteration 4: Log pseudolikelihood = -80678.234
```

Fitting full model:

```
tau = 0.0 Log pseudolikelihood = -80678.234
tau = 0.1 Log pseudolikelihood = -79721.835
tau = 0.2 Log pseudolikelihood = -78785.459
tau = 0.3 Log pseudolikelihood = -77887.322
tau = 0.4 Log pseudolikelihood = -77053.654
tau = 0.5 Log pseudolikelihood = -76325.354
tau = 0.6 Log pseudolikelihood = -75773.473
tau = 0.7 Log pseudolikelihood = -75539.749
tau = 0.8 Log pseudolikelihood = -75969.788
```

```
Iteration 0: Log pseudolikelihood = -75539.783
Iteration 1: Log pseudolikelihood = -74210.432
Iteration 2: Log pseudolikelihood = -74149.99
Iteration 3: Log pseudolikelihood = -74149.716
Iteration 4: Log pseudolikelihood = -74149.715
Iteration 5: Log pseudolikelihood = -74149.715
```

Calculating robust standard errors ...

Random-effects logistic regression
Group variable: **id**

Random effects u_i ~ Gaussian

Integration method: **mvaghermite**

Log pseudolikelihood = **-74149.715**

Number of obs = **164,331**
Number of groups = **58,731**

Obs per group:
min = **1**
avg = **2.8**
max = **4**

Integration pts. = **12**

Wald chi2(32) = **15579.98**
Prob > chi2 = **0.0000**

(Std. err. adjusted for 58,731

> clusters in id)

	fd	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.3776655	.0139859	-26.29	0.000	.3512
> 249						
> .4060966						
Above country average liabilities		.4114277	.0208233	-17.55	0.000	.3725
> 738						
> .4543335						
	mortg					
	No mortgage	1 (base)				
Mortgage		.8574957	.0333441	-3.95	0.000	.7945
> 707						
> .925404						
	thinc					
	No income	1 (base)				
Below country average income		.8768365	.0183273	-6.29	0.000	.8416
> 414						
> .9135033						
Above country average income		2.524492	.065177	35.87	0.000	2.399
> 925						
> 2.655524						
	single					
	No	1 (base)				
Yes		.4535342	.0131241	-27.32	0.000	.4285
> 272						
> .4800005						
	schanged					
Hooked up		.6551203	.040356	-6.87	0.000	.5806
> 123						
> .7391896						
	Unchanged	1 (base)				
Back single		1.46868	.096491	5.85	0.000	1.291
> 231						
> 1.670515						
	gali					
	Not limited	1 (base)				
Limited		.5024046	.0102587	-33.71	0.000	.4826
> 949						
> .522919						
	cjs					
	Retired	1 (base)				
Employed or self-employed		1.579599	.0538173	13.42	0.000	1.477
> 564						
> 1.688681						
	Unemployed	.2876609	.0204741	-17.51	0.000	.2502
> 057						
> .3307231						
	Permanently sick	.5035899	.0302666	-11.41	0.000	.4476
> 291						
> .5665467						
	Homemaker	.6284644	.0272384	-10.72	0.000	.5772
> 828						
> .6841838						
	Other	.7297371	.0518848	-4.43	0.000	.6348
> 125						
> .8388561						

> 408		age	1.055771	.0017186	33.34	0.000	1.052
>	1.059145						
> 331		yedu	1.124648	.0037452	35.27	0.000	1.117
>	1.132013						
> 467		gender					
>	1.118241	Male	1 (base)				
		Female	1.063515	.0272274	2.41	0.016	1.011
> 562		country					
>	.89627	Austria	1 (base)				
		Germany	.7644726	.0620387	-3.31	0.001	.6520
> 473		Sweden	.8933872	.0769282	-1.31	0.190	.7546
>	1.057634						
> 794		Spain	.1051147	.0075688	-31.29	0.000	.0912
>	.1210471						
> 794		Italy	.0362454	.002666	-45.10	0.000	.0313
>	.0418661						
> 599		France	.2084968	.0164885	-19.83	0.000	.1785
>	.2434528						
> 132		Denmark	1.664725	.1531075	5.54	0.000	1.390
>	1.993558						
> 014		Switzerland	1.053249	.0972456	0.56	0.574	.8789
>	1.262181						
> 341		Belgium	.2592305	.0195952	-17.86	0.000	.2235
>	.3006273						
> 941		Czechia	.1938851	.0144151	-22.06	0.000	.1675
>	.2243004						
> 797		Poland	.0308977	.002733	-39.31	0.000	.0259
>	.0367467						
> 051		Luxembourg	1.033075	.114282	0.29	0.769	.8317
>	1.283201						
> 378		Portugal	.0301476	.0027954	-37.76	0.000	.0251
>	.0361559						
> 266		Slovenia	.0350738	.0026383	-44.54	0.000	.030
>	.0406454						
> 847		Estonia	.0377091	.0027512	-44.93	0.000	.0326
>	.0435059						
> 137		Croatia	.0134982	.0011552	-50.30	0.000	.0114
>	.0159633						
> 094		_cons	.2532389	.0342243	-10.16	0.000	.1943
>	.3300404						
> 532		/lnsig2u	1.321209	.0197336			1.282
>	1.359886						

	sigma_u		1.935962	.0191018	1.898
> 883					
>	1.973765				
		rho		.5325443	.0049125
> 905					.522
>	.5421593				

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

```

194 * /// Mundlak_Gal_* Mundlak_sC_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Sin_* ///
>
195 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *)
> ctitle(xtlogitLim) stnum(replace coef=exp(coef), replace se=coef*se)
  results2.xls
  dir : seeout

196 asdoc, text(xtlogitLim) save(report.doc) append
  Click to Open File: report.doc

197 *asdoc test Mundlak_Gal_gali_Not Mundlak_Gal_gali_Limited Mundlak_sC_schanged_Hooked
> Mundlak_sC_schanged_Unchanged Mundlak_sC_schanged_Back Mundlak_Oth_cjs_Retired_Mund
> lak_Oth_cjs_Employed Mundlak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently_Mundlak_
> Oth_cjs_Homemaker Mundlak_Oth_cjs_Other Mundlak_Oth_liab_No_Mundlak_Oth_liab_Below_M
> undlak_Oth_liab_Above Mundlak_Oth_mortg_No_Mundlak_Oth_mortg_Mortgage Mundlak_Oth_th
> inc_No_Mundlak_Oth_thinc_Below Mundlak_Oth_thinc_Above Mundlak_OthC_age_Mundlak_OthC
> _yedu Mundlak_Sin_single_No_Mundlak_Sin_single_Yes, save(report.doc) append
198
199 xtlogit fd i.liab i.mortg i.thinc i.single ib1.schanged i.sphus i.cjs age yedu i.gen
> der i.country, or baselevels vce(robust)

```

Fitting comparison model:

```

Iteration 0: Log pseudolikelihood = -102965.85
Iteration 1: Log pseudolikelihood = -81332.109
Iteration 2: Log pseudolikelihood = -80297.411
Iteration 3: Log pseudolikelihood = -80289.905
Iteration 4: Log pseudolikelihood = -80289.901

```

Fitting full model:

```

tau = 0.0 Log pseudolikelihood = -80289.901
tau = 0.1 Log pseudolikelihood = -79344.767
tau = 0.2 Log pseudolikelihood = -78420.412
tau = 0.3 Log pseudolikelihood = -77535.33
tau = 0.4 Log pseudolikelihood = -76716.228
tau = 0.5 Log pseudolikelihood = -76004.762
tau = 0.6 Log pseudolikelihood = -75473.172
tau = 0.7 Log pseudolikelihood = -75265.093
tau = 0.8 Log pseudolikelihood = -75728.933

```

```

Iteration 0: Log pseudolikelihood = -75265.206
Iteration 1: Log pseudolikelihood = -73957.456
Iteration 2: Log pseudolikelihood = -73889.637
Iteration 3: Log pseudolikelihood = -73889.249
Iteration 4: Log pseudolikelihood = -73889.249
Iteration 5: Log pseudolikelihood = -73889.249

```

Calculating robust standard errors ...

Random-effects logistic regression
Group variable: **id**

Random effects u_i ~ Gaussian

Integration method: **mvaghermite**

Number of obs = **164,331**
Number of groups = **58,731**

Obs per group:
min = **1**
avg = **2.8**
max = **4**

Integration pts. = **12**

Log pseudolikelihood = **-73889.249**Wald chi2(35) = **15849.71**
Prob > chi2 = **0.0000**(Std. err. adjusted for **58,731**)> clusters in **id**)

	fd	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.3816923	.0140986	-26.08	0.000	.355
> 036						
> .4103498						
Above country average liabilities		.4071064	.0206187	-17.74	0.000	.3686
> 356						
> .4495921						
	mortg					
	No mortgage	1 (base)				
	Mortgage	.8437225	.0327478	-4.38	0.000	.7819
> 186						
> .9104115						
	thinc					
	No income	1 (base)				
Below country average income		.8685509	.0181488	-6.74	0.000	.8336
> 985						
> .9048602						
Above country average income		2.465921	.0636358	34.97	0.000	2.344
> 299						
> 2.593853						
	single					
	No	1 (base)				
	Yes	.4581193	.0131568	-27.18	0.000	.4330
> 447						
> .4846458						
	schanged					
	Hooked up	.6649296	.0408244	-6.65	0.000	.5895
> 421						
> .7499573						
	Unchanged	1 (base)				
	Back single	1.441004	.0945675	5.57	0.000	1.26
> 708						
> 1.638801						
	sphus					
	Excellent	1 (base)				
	Very good	.8676172	.0432214	-2.85	0.004	.7869
> 091						
> .956603						
	Good	.5886768	.0278024	-11.22	0.000	.536
> 631						
> .6457702						
	Fair	.368888	.0180803	-20.35	0.000	.3351
> 001						
> .4060827						
	Poor	.2082235	.0113622	-28.76	0.000	.1871
> 035						
> .2317275						
	cjs					
	Retired	1 (base)				
	Employed or self-employed	1.543866	.0524321	12.79	0.000	1.444
> 447						
> 1.650129						

> 363		Unemployed		.2961893	.0209564	-17.20	0.000	.2578
>	.3402473							
> 005		Permanently sick		.5570691	.0332313	-9.81	0.000	.4956
>	.6261617							
> 693		Homemaker		.635929	.0274343	-10.49	0.000	.5843
>	.6920379							
> 656		Other		.749065	.0531772	-4.07	0.000	.6517
>	.8608898							
> 793		age		1.05816	.0017208	34.76	0.000	1.054
>	1.061538							
> 831		yedu		1.115536	.0036992	32.97	0.000	1.10
>	1.12281							
> 793		gender						
>	1.098148	Male		1 (base)				
		Female		1.044713	.0265886	1.72	0.086	.9938
> 543		country						
>	.9866686	Austria		1 (base)				
		Germany		.8425922	.0678605	-2.13	0.033	.7195
> 647		Sweden		.8816562	.0756001	-1.47	0.142	.7452
>	1.043009							
> 971		Spain		.1254963	.0089666	-29.05	0.000	.1090
>	.1443604							
> 506		Italy		.0422887	.0030873	-43.33	0.000	.0366
>	.0487941							
> 744		France		.2343515	.0184307	-18.45	0.000	.2008
>	.2734077							
> 319		Denmark		1.61877	.1485379	5.25	0.000	1.352
>	1.937721							
> 302		Switzerland		1.031888	.0943856	0.34	0.731	.8625
>	1.2345							
> 746		Belgium		.263816	.0198316	-17.73	0.000	.2276
>	.3056945							
> 694		Czechia		.2104372	.0156058	-21.02	0.000	.1819
>	.2433585							
> 012		Poland		.0382546	.0033622	-37.13	0.000	.0322
>	.045446							
> 634		Luxembourg		1.114322	.1222262	0.99	0.324	.8987
>	1.38158							
> 973		Portugal		.0388072	.0035745	-35.28	0.000	.0323
>	.0464853							
> 622		Slovenia		.039094	.0029247	-43.33	0.000	.0337
>	.0452678							
> 401		Estonia		.0507608	.003696	-40.94	0.000	.04
>	.0585471							
> 738		Croatia		.0152103	.0012943	-49.19	0.000	.0128

>	.017971					
> 219		_cons	.3083588	.0432647	-8.39	0.000 .2342
>	.4059619					
> 095		/lnsig2u	1.300849	.0197728		1.262
>	1.339603					
> 578		sigma_u	1.916354	.0189458		1.879
>	1.953849					
> 043		rho	.5274726	.0049283		.5178
>	.5371203					

Note: Estimates are transformed only in the first equation to odds ratios.

Note: _cons estimates baseline odds (conditional on zero random effects).

```

200 * /// Mundlak_Sph_* Mundlak_sC_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Sin_* ///
>
201 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *)
> ctitle(xtlogitHea) stnum(replace coef=exp(coef), replace se=coef*se)
  results2.xls
  dir : seeout

202 asdoc, text(xtlogitHea) save(report.doc) append
  Click to Open File: report.doc

203 *asdoc test Mundlak_Sph_sphus_Excellent Mundlak_Sph_sphus_Very Mundlak_Sph_sphus_Goo
> d Mundlak_Sph_sphus_Fair Mundlak_Sph_sphus_Poor Mundlak_sC_schanged Hooked Mundlak_s
> C_schanged Unchanged Mundlak_sC_schanged Back Mundlak_Oth_cjs_Retired Mundlak_Oth_cj
> s_Employed Mundlak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_Oth_cjs_Ho
> memaker Mundlak_Oth_cjs_Other Mundlak_Oth_liab_No Mundlak_Oth_liab_Below Mundlak_Oth
> liab_Above Mundlak_Oth_mortg_No Mundlak_Oth_mortg_Mortgage Mundlak_Oth_thinc_No Mun
> dlak_Oth_thinc_Below Mundlak_Oth_thinc_Above Mundlak_OthC_age Mundlak_OthC_yedu Mund
> lak_Sin_single_No Mundlak_Sin_single_Yes, save(report.doc) append

204
205 xtologit fdistress i.liab i.mortg i.thinc i.single ibl.schanged i.gali i.cjs age yed
> u i.gender i.country [pweight=my_wgt], or baselevels vce(robust)

```

Fitting comparison model:

```

Iteration 0: Log likelihood = -4.654e+08
Iteration 1: Log likelihood = -4.148e+08
Iteration 2: Log likelihood = -3.826e+08
Iteration 3: Log likelihood = -3.757e+08
Iteration 4: Log likelihood = -3.755e+08
Iteration 5: Log likelihood = -3.755e+08
Iteration 6: Log likelihood = -3.755e+08

```

Refining starting values:

```

Grid node 0: Log likelihood = -3.613e+08

```

Fitting full model:

```

Iteration 0: Log pseudolikelihood = -3.613e+08
Iteration 1: Log pseudolikelihood = -3.507e+08
Iteration 2: Log pseudolikelihood = -3.500e+08
Iteration 3: Log pseudolikelihood = -3.500e+08
Iteration 4: Log pseudolikelihood = -3.500e+08
Iteration 5: Log pseudolikelihood = -3.500e+08

```

Random-effects ordered logistic regression
Group variable: **id**

Number of obs = **164,286**
Number of groups = **58,701**

Random effects u_i ~ Gaussian

Obs per group:
 min = 1
 avg = 2.8
 max = 4

Integration method: **mvaghermite**

Integration pts. = 12

Log pseudolikelihood = **-3.500e+08**

Wald chi2(32) = **9861.43**
 Prob > chi2 = **0.0000**

(Std. err. adjusted for 58,701

> clusters in id)

	fdistress	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
> 0388	Below country average liabilities	.3815563	.0218547	-16.82	0.000	.341
>	.4268876					
>	Above country average liabilities	.3886611	.0285518	-12.86	0.000	.336
> 5426						
>	.4488509					
	mortg					
	No mortgage	1 (base)				
> 7053	Mortgage	.7279161	.0410736	-5.63	0.000	.651
>	.8130391					
	thinc					
	No income	1 (base)				
> 1319	Below country average income	.9063044	.0301309	-2.96	0.003	.849
>	.9673264					
>	Above country average income	2.541774	.1021432	23.21	0.000	2.34
> 9259						
>	2.750067					
	single					
	No	1 (base)				
> 1509	Yes	.5696717	.0269918	-11.88	0.000	.519
>	.6251089					
	schanged					
> 8406	Hooked up	.7510662	.0705085	-3.05	0.002	.624
>	.9027908					
	Unchanged	1 (base)				
> 8593	Back single	1.267047	.1411016	2.13	0.034	1.01
>	1.576104					
	gali					
	Not limited	1 (base)				
> 9474	Limited	.6127591	.0182859	-16.41	0.000	.577
>	.6496676					
	cjs					
	Retired	1 (base)				
> 7782	Employed or self-employed	1.231962	.0611349	4.20	0.000	1.11
>	1.357805					
> 4033	Unemployed	.1857747	.0225748	-13.85	0.000	.146

>	.2357341	Permanently sick		.4345991	.0416254	-8.70	0.000	.360
> 2149								
>	.5243437	Homemaker		.6594184	.0444488	-6.18	0.000	.577
> 8098								
>	.7525532	Other		.5635106	.0701088	-4.61	0.000	.441
> 5708								
>	.7191241							
		age		1.050811	.0027539	18.91	0.000	1.04
> 5427								
>	1.056222	yedu		1.128013	.0064134	21.19	0.000	1.11
> 5513								
>	1.140653							
		gender						
		Male		1	(base)			
> 4658		Female		1.069844	.0442898	1.63	0.103	.986
>	1.160269							
		country						
		Austria		1	(base)			
> 1535		Germany		.6530935	.0559452	-4.97	0.000	.552
>	.7724865							
> 7906		Sweden		1.164882	.110166	1.61	0.107	.967
>	1.40211							
> 8277		Spain		.1137886	.0124489	-19.87	0.000	.091
>	.1410014							
> 2277		Italy		.0369348	.0031631	-38.52	0.000	.031
>	.043685							
> 4717		France		.2222739	.0193112	-17.31	0.000	.187
>	.2635367							
> 4166		Denmark		2.696831	.2528239	10.58	0.000	2.24
>	3.240803							
> 4327		Switzerland		1.235454	.1181285	2.21	0.027	1.02
>	1.490097							
> 0105		Belgium		.4061395	.0368268	-9.94	0.000	.340
>	.48513							
> 8506		Czechia		.1575828	.015563	-18.71	0.000	.129
>	.1912379							
> 1427		Poland		.0438329	.0043142	-31.77	0.000	.036
>	.0531593							
> 8972		Luxembourg		.6910535	.0818454	-3.12	0.002	.547
>	.8716142							
> 0811		Portugal		.0363631	.0061659	-19.55	0.000	.026
>	.0506986							
> 4695		Slovenia		.0556874	.0048491	-33.17	0.000	.0
>	.0660507							
> 4227		Estonia		.0465876	.0039693	-35.99	0.000	.039
>	.0550546							
> 9304		Croatia		.0230828	.0021054	-41.32	0.000	.01
>	.0276012							

		/cut1	-1.456908	.2088315	-1.86
> 6211					
>	-1.047606	/cut2	1.344979	.209017	.935
> 3131					
>	1.754645	/cut3	4.001567	.2129217	3.58
> 4248					
>	4.418886				
		/sigma2_u	3.408337	.1077956	3.20
> 3476					
>	3.626298				

Note: Estimates are transformed only in the first equation to odds ratios.

```

206 * /// Mundlak_Gal_* Mundlak_sC_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Sin_* ///
>
207 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(**, **, *)
> ctitle(xtologitLim) stnum(replace coef=exp(coef), replace se=coef*se)
  results2.xls
  dir : seeout
208 asdoc, text(xtologitLim) save(report.doc) append
  Click to Open File: report.doc
209 *asdoc test Mundlak_Gal_gali_Not Mundlak_Gal_gali_Limited Mundlak_sC_schanged_Hooked
> Mundlak_sC_schanged_Unchanged Mundlak_sC_schanged_Back Mundlak_Oth_cjs_Retired Mund
> lak_Oth_cjs_Employed Mundlak_Oth_cjs_Unemployed Mundlak_Oth_cjs_Permanently Mundlak_
> Oth_cjs_Homemaker Mundlak_Oth_cjs_Other Mundlak_Oth_liab_No Mundlak_Oth_liab_Below M
> undlak_Oth_liab_Above Mundlak_Oth_mortg_No Mundlak_Oth_mortg_Mortgage Mundlak_Oth_th
> inc_No Mundlak_Oth_thinc_Below Mundlak_Oth_thinc_Above Mundlak_OthC_age Mundlak_OthC
> _yed_u Mundlak_Sin_single_No Mundlak_Sin_single_Yes, save(report.doc) append
210
211 xtologit fdistress i.liab i.mortg i.thinc i.single ib1.schanged i.sphus i.cjs age ye
> du i.gender i.country [pweight=my_wgt], or baselevels vce(robust)

```

Fitting comparison model:

```

Iteration 0: Log likelihood = -4.654e+08
Iteration 1: Log likelihood = -4.128e+08
Iteration 2: Log likelihood = -3.800e+08
Iteration 3: Log likelihood = -3.731e+08
Iteration 4: Log likelihood = -3.730e+08
Iteration 5: Log likelihood = -3.730e+08
Iteration 6: Log likelihood = -3.730e+08

```

Refining starting values:

```
Grid node 0: Log likelihood = -3.592e+08
```

Fitting full model:

```

Iteration 0: Log pseudolikelihood = -3.592e+08
Iteration 1: Log pseudolikelihood = -3.492e+08
Iteration 2: Log pseudolikelihood = -3.486e+08
Iteration 3: Log pseudolikelihood = -3.486e+08
Iteration 4: Log pseudolikelihood = -3.486e+08
Iteration 5: Log pseudolikelihood = -3.486e+08

```

Random-effects ordered logistic regression
Group variable: **id**

Random effects u_i ~ Gaussian

Number of obs = **164,286**
Number of groups = **58,701**

Obs per group:
min = **1**
avg = **2.8**
max = **4**

Integration method: **mvaghermite**Integration pts. = **12**Log pseudolikelihood = **-3.486e+08**Wald chi2(35) = **10270.44**
Prob > chi2 = **0.0000**(Std. err. adjusted for **58,701**)> clusters in **id**)

	fdistress	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.3853396	.0218646	-16.81	0.000	.344
> 7828						
> .4306671						
Above country average liabilities		.3942787	.0292737	-12.54	0.000	.340
> 8826						
> .4560388						
	mortg					
	No mortgage	1 (base)				
Mortgage		.727359	.0412622	-5.61	0.000	.650
> 8205						
> .8128987						
	thinc					
	No income	1 (base)				
Below country average income		.8932368	.0295752	-3.41	0.001	.837
> 1112						
> .9531253						
Above country average income		2.48282	.1003237	22.51	0.000	2.29
> 3774						
> 2.687447						
	single					
	No	1 (base)				
Yes		.5822615	.0270323	-11.65	0.000	.531
> 6182						
> .6377293						
	schanged					
	Hooked up	.7692381	.0711951	-2.83	0.005	.641
> 6228						
> .9222354						
	Unchanged	1 (base)				
Back single		1.248978	.1353606	2.05	0.040	1.00
> 9959						
> 1.544562						
	sphus					
	Excellent	1 (base)				
Very good		.9639897	.0832406	-0.42	0.671	.813
> 9001						
> 1.141757						
	Good	.6581393	.0532518	-5.17	0.000	.56
> 1623						
> .7712423						
	Fair	.4479736	.0370306	-9.71	0.000	.380
> 9693						
> .5267625						
	Poor	.2533645	.0234855	-14.81	0.000	.21
> 1273						
> .3038417						
	cjs					
	Retired	1 (base)				
Employed or self-employed		1.199606	.0587255	3.72	0.000	1.08

> 9855							
>	1.320408						
		Unemployed		.1920213	.0233832	-13.55	0.000 .151
> 2499							
>	.2437832						
		Permanently sick		.4988259	.0472368	-7.34	0.000 .414
> 3273							
>	.6005572						
		Homemaker		.6656004	.0444042	-6.10	0.000 .584
> 0196							
>	.7585772						
		Other		.5801835	.0715873	-4.41	0.000 .45
> 5552							
>	.7389122						
		age		1.054262	.0027265	20.43	0.000 1.04
> 8932							
>	1.05962						
		yedu		1.120315	.0062586	20.34	0.000 1.10
> 8115							
>	1.132649						
		gender					
		Male		1	(base)		
		Female		1.060012	.043088	1.43	0.152 .97
> 8838							
>	1.147918						
		country					
		Austria		1	(base)		
		Germany		.7209089	.060434	-3.90	0.000 .611
> 6794							
>	.8496438						
		Sweden		1.138244	.105722	1.39	0.163 .948
> 7989							
>	1.365514						
		Spain		.1318968	.0141745	-18.85	0.000 .10
> 6846							
>	.1628209						
		Italy		.0420078	.003523	-37.80	0.000 .035
> 6405							
>	.0495126						
		France		.2482812	.021143	-16.36	0.000 .210
> 1153							
>	.2933797						
		Denmark		2.58758	.2391143	10.29	0.000 2.15
> 8915							
>	3.101359						
		Switzerland		1.196699	.1119401	1.92	0.055 .996
> 2377							
>	1.437497						
		Belgium		.4124086	.0366205	-9.97	0.000 .346
> 5323							
>	.490808						
		Czechia		.170941	.0165231	-18.27	0.000 .141
> 4391							
>	.2065966						
		Poland		.0531401	.0051427	-30.33	0.000 .043
> 9588							
>	.0642389						
		Luxembourg		.7417273	.0864485	-2.56	0.010 .590
> 2506							
>	.9320776						
		Portugal		.0461409	.0075603	-18.77	0.000 .033
> 4667							
>	.0636148						
		Slovenia		.0614321	.0052447	-32.68	0.000 .051
> 9667							
>	.0726216						
		Estonia		.0614807	.0051803	-33.10	0.000 .052
> 1215							
>	.0725205						

	Croatia		.0260009	.0023292	-40.74	0.000	.021
> 8141							
>	.0309913						
	/cut1		-1.528575	.220504			-1.96
> 0755							
>	-1.096395						
	/cut2		1.274068	.2200851			.842
> 7091							
>	1.705427						
	/cut3		3.927405	.2233067			3.48
> 9732							
>	4.365078						
	/sigma2_u		3.26379	.1037095			3.06
> 6723							
>	3.47352						

Note: Estimates are transformed only in the first equation to odds ratios.

```

212 * /// Mundlak_Sph_* Mundlak_sC_* Mundlak_Oth_* Mundlak_OthC_* Mundlak_Sin_* ///
>
213 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(xtologitHea) stnum(replace coef=exp(coef), replace se=coef*se)
  results2.xls
  dir : seeout
214 asdoc, text(xtologitHea) save(report.doc) append
  Click to Open File: report.doc
215 *asdoc test Mundlak_Sph sphus Excellent Mundlak_Sph sphus Very Mundlak_Sph sphus Goo
> d Mundlak_Sph sphus Fair Mundlak_Sph sphus Poor Mundlak_sC_schanged Hooked Mundlak_s
> C_schanged Unchanged Mundlak_sC_schanged Back Mundlak_Oth_cjs Retired Mundlak_Oth_cj
> s_Employed Mundlak_Oth_cjs Unemployed Mundlak_Oth_cjs Permanently Mundlak_Oth_cjs_Ho
> memaker Mundlak_Oth_cjs_Other Mundlak_Oth_liab No Mundlak_Oth_liab Below Mundlak_Oth
> liab Above Mundlak_Oth_mortg No Mundlak_Oth_mortg Mortgage Mundlak_Oth_thinc No Mun
> dlak_Oth_thinc Below Mundlak_Oth_thinc Above Mundlak_OthC_age Mundlak_OthC_yedu Mund
> lak_Sin_single No Mundlak_Sin_single Yes, save(report.doc) append
216
217 logit fd i.liab i.mortg i.thinc i.single ibl.schanged i.gali i.cjs age yedu i.gender
> i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)

```

```

Iteration 0: Log pseudolikelihood = -2.184e+08
Iteration 1: Log pseudolikelihood = -1.698e+08
Iteration 2: Log pseudolikelihood = -1.680e+08
Iteration 3: Log pseudolikelihood = -1.680e+08
Iteration 4: Log pseudolikelihood = -1.680e+08

```

Logistic regression

```

Number of obs = 164,286
Wald chi2(35) = 8564.14
Prob > chi2   = 0.0000
Pseudo R2    = 0.2310

```

Log pseudolikelihood = -1.680e+08

(Std. err. adjusted for 58,701

> clusters in id)

	fd	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1	(base)			
Below country average liabilities		.3658071	.0185127	-19.87	0.000	.3312
> 643						
>	.4039519					
Above country average liabilities		.3384315	.0264338	-13.87	0.000	.290

> 393							
>	.3944168						
		mortg					
		No mortgage	1	(base)			
		Mortgage	.7605482	.0470143	-4.43	0.000	.6737
> 652							
>	.8585091						
		thinc					
		No income	1	(base)			
		Below country average income	1.029373	.0456898	0.65	0.514	.9436
> 076							
>	1.122934						
		Above country average income	3.383523	.1702669	24.22	0.000	3.065
> 735							
>	3.734251						
		single					
		No	1	(base)			
		Yes	.7269293	.028062	-8.26	0.000	.6739
> 581							
>	.7840639						
		schanged					
		Hooked up	.797885	.0785102	-2.29	0.022	.6579
> 363							
>	.967602						
		Unchanged	1	(base)			
		Back single	1.111528	.1002581	1.17	0.241	.931
> 415							
>	1.32647						
		gali					
		Not limited	1	(base)			
		Limited	.5595237	.016095	-20.19	0.000	.5288
> 509							
>	.5919756						
		cjs					
		Retired	1	(base)			
		Employed or self-employed	1.137909	.0556257	2.64	0.008	1.033
> 945							
>	1.252327						
		Unemployed	.2128135	.0222174	-14.82	0.000	.1734
> 342							
>	.261134						
		Permanently sick	.4804059	.0399783	-8.81	0.000	.4081
> 062							
>	.5655141						
		Homemaker	.7362774	.0415197	-5.43	0.000	.6592
> 362							
>	.8223219						
		Other	.6677751	.0679678	-3.97	0.000	.5470
> 069							
>	.8152066						
		age	1.033308	.0023224	14.58	0.000	1.028
> 766							
>	1.03787						
		yedu	1.073952	.004908	15.61	0.000	1.064
> 376							
>	1.083615						
		gender					
		Male	1	(base)			
		Female	1.048406	.0370826	1.34	0.181	.9781
> 871							
>	1.123665						
		country					
		Austria	1	(base)			

> 029		Germany		.804306	.064452	-2.72	0.007	.6874
>	.9410901							
> 028		Sweden		1.137244	.1068166	1.37	0.171	.946
>	1.36711							
> 138		Spain		.1950271	.0172938	-18.43	0.000	.1639
>	.2320461							
> 872		Italy		.0998238	.0075402	-30.51	0.000	.0860
>	.1157523							
> 266		France		.3509265	.0275214	-13.35	0.000	.3009
>	.409234							
> 274		Denmark		1.606889	.1472362	5.18	0.000	1.34
>	1.923002							
> 615		Switzerland		1.024868	.0960074	0.26	0.793	.8529
>	1.231421							
> 288		Belgium		.4741515	.037701	-9.39	0.000	.4057
>	.554113							
> 753		Czechia		.3082859	.0297402	-12.20	0.000	.2551
>	.3724506							
> 165		Poland		.1028359	.0088245	-26.51	0.000	.0869
>	.121671							
> 758		Luxembourg		.9150841	.1056468	-0.77	0.442	.7297
>	1.147447							
> 655		Portugal		.0984535	.0125647	-18.16	0.000	.0766
>	.1264335							
> 921		Slovenia		.1108545	.0084373	-28.90	0.000	.0954
>	.1286883							
> 469		Estonia		.1261503	.0094366	-27.68	0.000	.1089
>	.1460702							
> 746		Croatia		.0625717	.0050156	-34.57	0.000	.0534
>	.0732165							
> 058		Qyear 2015			1 (base)			
>	1.368911	2017		1.275819	.0458439	6.78	0.000	1.189
> 043		2020		1.156048	.040112	4.18	0.000	1.080
>	1.2374							
> 214		2022		1.279379	.0471563	6.68	0.000	1.190
>	1.375224							
> 038		_cons		.4212949	.0756513	-4.81	0.000	.2963
>	.5990114							

Note: **_cons** estimates baseline odds.

```

218 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(logitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout

```

219

```

220 logit fd i.liab i.mortg i.thinc i.single ibl.schanged i.sphus i.cjs age yedu i.gende
> r i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)

```

```

Iteration 0: Log pseudolikelihood = -2.184e+08
Iteration 1: Log pseudolikelihood = -1.684e+08
Iteration 2: Log pseudolikelihood = -1.665e+08
Iteration 3: Log pseudolikelihood = -1.664e+08
Iteration 4: Log pseudolikelihood = -1.664e+08

```

Logistic regression

```

Number of obs = 164,286
Wald chi2(38) = 8705.56
Prob > chi2   = 0.0000
Pseudo R2    = 0.2380

```

Log pseudolikelihood = -1.664e+08

(Std. err. adjusted for 58,701

> clusters in id)

	fd	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.3726232	.0187258	-19.64	0.000	.3376
> 708						
> .4111934						
Above country average liabilities		.3428614	.0272782	-13.45	0.000	.2933
> 571						
> .4007196						
	mortg					
	No mortgage	1 (base)				
	Mortgage	.753372	.0474129	-4.50	0.000	.665
> 947						
> .8522739						
	thinc					
	No income	1 (base)				
Below country average income		1.031303	.0457313	0.70	0.487	.9454
> 557						
> 1.124945						
Above country average income		3.350809	.1697026	23.88	0.000	3.034
> 173						
> 3.700488						
	single					
	No	1 (base)				
	Yes	.7352068	.0281542	-8.03	0.000	.6820
> 456						
> .7925116						
	schanged					
	Hooked up	.8170889	.0791778	-2.08	0.037	.6757
> 498						
> .9879904						
	Unchanged	1 (base)				
	Back single	1.112845	.0994423	1.20	0.231	.9340
> 554						
> 1.325858						
	sphus					
	Excellent	1 (base)				
	Very good	1.073296	.0997115	0.76	0.446	.8946

> 246							
>	1.28765						
> 861		Good		.7210276	.0615999	-3.83	0.000 .609
>	.852458						
> 875		Fair		.470919	.0412151	-8.60	0.000 .3966
>	.5590413						
> 423		Poor		.2874326	.0266674	-13.44	0.000 .2396
>	.3447535						
		cjs					
		Retired		1	(base)		
> 339	Employed or self-employed			1.103486	.0541273	2.01	0.045 1.002
>	1.214841						
> 788		Unemployed		.2190457	.0227645	-14.61	0.000 .1786
>	.2685321						
> 502		Permanently sick		.5519978	.0453202	-7.24	0.000 .4699
>	.6483699						
> 207		Homemaker		.7375428	.0412769	-5.44	0.000 .6609
>	.8230479						
> 402		Other		.6759882	.0692366	-3.82	0.000 .5530
>	.8262693						
		age		1.036368	.0023312	15.88	0.000 1.031
> 809							
>	1.040947						
		yedu		1.069134	.0048661	14.69	0.000 1.059
> 639							
>	1.078714						
		gender					
		Male		1	(base)		
> 034		Female		1.040699	.0366969	1.13	0.258 .9712
>	1.115167						
		country					
		Austria		1	(base)		
> 964		Germany		.8761829	.0692793	-1.67	0.095 .7503
>	1.023054						
> 894		Sweden		1.12897	.1049707	1.30	0.192 .9408
>	1.354648						
> 978		Spain		.2229419	.0196948	-16.99	0.000 .1874
>	.2650864						
> 633		Italy		.1118483	.0083266	-29.43	0.000 .0966
>	.1294188						
> 595		France		.3858732	.0299854	-12.25	0.000 .3313
>	.4493552						
> 182		Denmark		1.576001	.1436305	4.99	0.000 1.3
>	1.88422						
> 732		Switzerland		1.003639	.0931122	0.04	0.969 .8367
>	1.203781						
> 762		Belgium		.4720085	.0370143	-9.57	0.000 .404
>	.5504273						
> 382		Czechia		.32168	.0309237	-11.80	0.000 .2664
>	.3883753						

> 509		Poland		.1191062	.0101707	-24.92	0.000	.1007
>	.1408057							
> 178		Luxembourg		.9774971	.1124892	-0.20	0.843	.7801
>	1.224816							
> 863		Portugal		.1216672	.015448	-16.59	0.000	.094
>	.156045							
> 193		Slovenia		.1182525	.0089089	-28.34	0.000	.1020
>	.1370686							
> 026		Estonia		.1607505	.0119811	-24.52	0.000	.1389
>	.1860349							
> 777		Croatia		.0675369	.0053825	-33.82	0.000	.05
>	.0789551							
		Qyear						
		2015		1	(base)			
> 889		2017		1.276076	.0460769	6.75	0.000	1.188
>	1.369658							
> 032		2020		1.123087	.0396336	3.29	0.001	1.048
>	1.203517							
> 911		2022		1.249282	.0462132	6.02	0.000	1.161
>	1.343222							
		_cons		.4087693	.079623	-4.59	0.000	.2790
> 456								
>	.5987994							

Note: **_cons** estimates baseline odds.

```
221 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(logitHea) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout
```

222

```
223 ologit fdistress i.liab i.mortg i.thinc i.single ibl.schanged i.gali i.cjs age yedu
> i.gender i.country i.Qyear
```

```
Iteration 0: Log likelihood = -210570.75
Iteration 1: Log likelihood = -180324.76
Iteration 2: Log likelihood = -179444.66
Iteration 3: Log likelihood = -179443.29
Iteration 4: Log likelihood = -179443.29
```

Ordered logistic regression

```
Number of obs = 164,331
LR chi2(35) = 62254.92
Prob > chi2 = 0.0000
Pseudo R2 = 0.1478
```

Log likelihood = -179443.29

	fdistress	Coefficient	Std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
Below country average liabilities		-.8255526	.020122	-41.03	0.000	-.864
> 991						
>		-.7861143				
Above country average liabilities		-.6635903	.0272139	-24.38	0.000	-.7169
> 285						
>		-.610252				

224 asdoc estat parallel, save(report.doc) append

Tests of proportional odds assumption

Number of obs = **164,331**
 Number of predictors = **35**
 Number of outcome levels = **4**
 Degrees of freedom = **70**

Test	chi2	P>chi2
Brant	3034.25	0.000
Likelihood-ratio	3049.48	0.000
Score	3105.28	0.000
Wald	3148.43	0.000
Wolfe-Gould	3060.09	0.000

Click to Open File: [report.doc](#)

225

226 ologit fdistress i.liab i.mortg i.thinc i.single ibl.schanged i.gali i.cjs age yedu
 > i.gender i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)

Iteration 0: Log pseudolikelihood = **-4.452e+08**
 Iteration 1: Log pseudolikelihood = **-3.776e+08**
 Iteration 2: Log pseudolikelihood = **-3.753e+08**
 Iteration 3: Log pseudolikelihood = **-3.753e+08**
 Iteration 4: Log pseudolikelihood = **-3.753e+08**

Ordered logistic regression

Number of obs = **164,286**
 Wald chi2(35) = **13634.95**
 Prob > chi2 = **0.0000**
 Pseudo R2 = **0.1570**

Log pseudolikelihood = **-3.753e+08**

(Std. err. adjusted for **58,701**)

> clusters in **id**)

	fdistress	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.3729123	.0161653	-22.76	0.000	.3425
> 375						
> .4059806						
Above country average liabilities		.3993685	.0240032	-15.27	0.000	.3549
> 884						
> .449297						
	mortg					
	No mortgage	1 (base)				
Mortgage		.7674726	.0341741	-5.94	0.000	.7033
> 321						
> .8374623						
	thinc					
	No income	1 (base)				
Below country average income		1.028624	.0365319	0.79	0.427	.9594
> 577						
> 1.102776						
Above country average income		3.210239	.1246108	30.05	0.000	2.975
> 065						
> 3.464002						
	single					
	No	1 (base)				
Yes		.7790523	.0253555	-7.67	0.000	.7309
> 084						
> .8303674						

> 929		schanged Hooked up	.8080364	.0592138	-2.91	0.004	.699
>	.9328415						
> 763		Unchanged Back single	1 1.137765	(base) .0911246	1.61	0.107	.9724
>	1.331147						
> 306		gali Not limited Limited	1 .6050751	(base) .0134359	-22.63	0.000	.579
>	.6319904						
> 481		cjs Retired Employed or self-employed	1 1.131836	(base) .0419777	3.34	0.001	1.052
>	1.217175						
> 506		Unemployed	.2035847	.0190626	-17.00	0.000	.1694
>	.2445947						
> 702		Permanently sick	.4626952	.0328612	-10.85	0.000	.4025
>	.5318						
> 013		Homemaker	.7416607	.036086	-6.14	0.000	.6742
>	.8158701						
> 124		Other	.6342095	.0588097	-4.91	0.000	.5288
>	.7606132						
> 472		age	1.030131	.0018701	16.35	0.000	1.026
>	1.033803						
> 313		yedu	1.076293	.0040866	19.36	0.000	1.068
>	1.084333						
> 211		gender Male Female	1 1.043021	(base) .0284517	1.54	0.123	.9887
>	1.100303						
> 784		country Austria Germany	1 .7968324	(base) .0454342	-3.98	0.000	.7125
>	.8910483						
> 466		Sweden	1.145865	.0740957	2.11	0.035	1.009
>	1.300695						
> 407		Spain	.2300048	.0168501	-20.06	0.000	.1992
>	.2655191						
> 663		Italy	.10331	.0057312	-40.92	0.000	.0926
>	.1151764						
> 381		France	.3807616	.0220893	-16.64	0.000	.3398
>	.4266131						
> 445		Denmark	2.108262	.1362456	11.54	0.000	1.857
>	2.392947						
> 996		Switzerland	1.065796	.0694561	0.98	0.328	.9379
>	1.211003						
> 996		Belgium	.5990315	.0362125	-8.48	0.000	.5320

>	.6743826						
> 285		Czechia		.2981376	.0198706	-18.16	0.000 .2616
>	.3397415						
> 983		Poland		.1157406	.0074061	-33.70	0.000 .1020
>	.1312059						
> 347		Luxembourg		.8500839	.0674001	-2.05	0.041 .7277
>	.9930028						
> 949		Portugal		.109576	.0113177	-21.41	0.000 .0894
>	.1341629						
> 927		Slovenia		.1413394	.0081172	-34.07	0.000 .1262
>	.1581789						
> 568		Estonia		.1325375	.0073672	-36.36	0.000 .1188
>	.1477929						
> 576		Croatia		.0805604	.0047562	-42.66	0.000 .0717
>	.0904431						
		Qyear					
		2015		1	(base)		
> 515		2017		1.239991	.0348599	7.65	0.000 1.173
>	1.310232						
> 166		2020		1.000643	.024817	0.03	0.979 .953
>	1.050485						
> 463		2022		1.08019	.0286547	2.91	0.004 1.025
>	1.137838						
		/cut1		-1.182113	.1470668		-1.470
> 358							
>	-.893867						
		/cut2		.7619762	.1461576		.4755
> 125							
>	1.04844						
		/cut3		2.549639	.1469925		2.261
> 539							
>	2.837738						

Note: Estimates are transformed only in the first equation to odds ratios.

```
227 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(ologitLim) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout
```

228

```
229 ologit fdistress i.liab i.mortg i.thinc i.single ibl.schanged i.sphus i.cjs age yedu
> i.gender i.country i.Qyear [pweight=my_wgt], or baselevels vce(cluster id)
```

```
Iteration 0: Log pseudolikelihood = -4.452e+08
Iteration 1: Log pseudolikelihood = -3.752e+08
Iteration 2: Log pseudolikelihood = -3.728e+08
Iteration 3: Log pseudolikelihood = -3.728e+08
Iteration 4: Log pseudolikelihood = -3.728e+08
```

Ordered logistic regression

Log pseudolikelihood = -3.728e+08

```
Number of obs = 164,286
Wald chi2(38) = 14196.41
Prob > chi2 = 0.0000
Pseudo R2 = 0.1626
```

(Std. err. adjusted for 58,701

> clusters in id)

	fdistress	Odds ratio	Robust std. err.	z	P> z	[95%
> con						
> f. interval]						
	liab					
	No liabilities	1 (base)				
Below country average liabilities		.3802355	.016279	-22.59	0.000	.3496
> 313						
> .4135186						
Above country average liabilities		.4062461	.0247349	-14.79	0.000	.3605
> 476						
> .4577368						
	mortg					
	No mortgage	1 (base)				
Mortgage		.7607425	.0340078	-6.12	0.000	.696
> 925						
> .8304038						
	thinc					
	No income	1 (base)				
Below country average income		1.029226	.0367494	0.81	0.420	.9596
> 609						
> 1.103834						
Above country average income		3.167843	.1237436	29.52	0.000	2.934
> 362						
> 3.419902						
	single					
	No	1 (base)				
Yes		.7847026	.0252267	-7.54	0.000	.7367
> 846						
> .835737						
	schanged					
Hooked up		.8265643	.0597858	-2.63	0.008	.7173
> 131						
> .9524551						
	Unchanged	1 (base)				
Back single		1.143478	.0896979	1.71	0.087	.9805
> 215						
> 1.333518						
	sphus					
Excellent		1 (base)				
Very good		.9534473	.0644941	-0.70	0.481	.8350
> 622						
> 1.088616						
	Good	.6735687	.0436308	-6.10	0.000	.5932
> 596						
> .7647491						
	Fair	.462159	.030853	-11.56	0.000	.4054
> 773						
> .5267643						
	Poor	.2780692	.020232	-17.59	0.000	.241
> 113						
> .3206899						
	cjs					
	Retired	1 (base)				
Employed or self-employed		1.100032	.04079	2.57	0.010	1.022
> 921						
> 1.182956						
	Unemployed	.2101593	.0197915	-16.56	0.000	.1747
> 381						
> .2527608						

> 295		Permanently sick		.538356	.0373973	-8.91	0.000	.4698
>	.6168773							
> 613		Homemaker		.7442173	.0359096	-6.12	0.000	.6770
>	.8180344							
> 837		Other		.6430179	.0588766	-4.82	0.000	.5373
>	.7694167							
> 669		age		1.033304	.001858	18.22	0.000	1.029
>	1.036952							
> 576		yedu		1.071422	.004018	18.40	0.000	1.063
>	1.079326							
> 563		gender						
>	1.091139	Male		1	(base)			
		Female		1.03474	.0280186	1.26	0.207	.9812
> 107		country						
>	.9664617	Austria		1	(base)			
		Germany		.8652908	.0488174	-2.56	0.010	.7747
> 733		Sweden		1.130803	.072788	1.91	0.056	.9967
>	1.282854							
> 245		Spain		.2614938	.0190353	-18.43	0.000	.2267
>	.3015951							
> 178		Italy		.1148367	.0063067	-39.41	0.000	.1031
>	.1278874							
> 684		France		.4156387	.0239402	-15.24	0.000	.3712
>	.4653116							
> 409		Denmark		2.055863	.132785	11.16	0.000	1.811
>	2.333308							
> 269		Switzerland		1.049556	.0676912	0.75	0.453	.9249
>	1.190979							
> 676		Belgium		.6002346	.0359938	-8.51	0.000	.533
>	.6750943							
> 342		Czechia		.3121352	.0206743	-17.58	0.000	.2741
>	.3554039							
> 163		Poland		.13354	.0085354	-31.50	0.000	.1178
>	.1513621							
> 044		Luxembourg		.8996131	.0711455	-1.34	0.181	.77
>	1.050444							
> 702		Portugal		.1340579	.0136093	-19.79	0.000	.1098
>	.1635704							
> 114		Slovenia		.1510662	.0086031	-33.19	0.000	.1351
>	.1689051							
> 498		Estonia		.167087	.009339	-32.01	0.000	.1497
>	.1864315							
> 131		Croatia		.0875254	.0051378	-41.50	0.000	.0780
>	.0981975							
		Qyear						

		2015		1	(base)			
> 635		2017		1.241582	.0351126	7.65	0.000	1.174
>	1.312344							
> 437		2020		.9770682	.0244284	-0.93	0.353	.9303
>	1.026139							
> 647		2022		1.057501	.0282016	2.10	0.036	1.003
>	1.114245							
> 629		/cut1		-1.25818	.1589056			-1.569
>	-.9467307							
> 902		/cut2		.7059832	.1576974			.396
>	1.015064							
> 848		/cut3		2.51167	.1580753			2.201
>	2.821492							

Note: Estimates are transformed only in the first equation to odds ratios.

```
230 outreg2 using results2.xls, excel dec(3) alpha(0.01, 0.05, 0.10) symbol(***, **, *)
> ctitle(ologitHea) stnum(replace coef=exp(coef), replace se=coef*se)
results2.xls
dir : seeout
```

```
231
232
233 log close DistressAndMarriage
      name: DistressAndMarriage
      log:  C:\Users\hansg\Documents\working\SHARE data 4\output.smcl
      log type: smcl
closed on: 30 Jun 2026, 11:34:11
```
